DBMS MINIPROJECT

GROUP- G22
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PROJECT TITLE:Regulatory Compliance Validation and Violation Explanation for Airline Crew Rosters

PART A : SCHEMA (DDL) -

A0 - Create and connect to the submission database

```
PS C:\Users\cshri> & "C:\Program Files\PostgreSQL\17\bin\psql.exe" -U postgres -h localhost -p 5432 -d postgres
psql (17.11)
WARNING: Console code page (437) differs from Windows code page (1252)
         8-bit characters might not work correctly. See psql reference
         page "Notes for Windows users" for details.
Type "help" for help.

postgres=# \set PROMPT1 'PES2UG24CS436 | PES2UG24CS495 | %/%# '
PES2UG24CS436 | PES2UG24CS495 | postgres# CREATE DATABASE crewguard_submission;
CREATE DATABASE
PES2UG24CS436 | PES2UG24CS495 | postgres# \c crewguard_submission
You are now connected to database "crewguard_submission" as user "postgres".
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

A1 - Extension and first seven tables (incl. exclusion constraint)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE EXTENSION IF NOT EXISTS btree_gist;
CREATE EXTENSION
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS airport (
crewguard_submission(#     iata_code       CHAR(3)      PRIMARY KEY,
crewguard_submission(#     name            VARCHAR(120) NOT NULL,
crewguard_submission(#     city            VARCHAR(80)  NOT NULL,
crewguard_submission(#     country         VARCHAR(80)  NOT NULL,
crewguard_submission(#     timezone        VARCHAR(64)  NOT NULL,
crewguard_submission(#     CONSTRAINT chk_airport_iata_format CHECK (iata_code ~ '^[A-Z]{3}$')
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS aircraft_type (
crewguard_submission(#     type_code       VARCHAR(10)  PRIMARY KEY,
crewguard_submission(#     manufacturer    VARCHAR(60)  NOT NULL,
crewguard_submission(#     description     VARCHAR(160),
crewguard_submission(#     CONSTRAINT chk_aircraft_type_upper CHECK (type_code = UPPER(type_code))
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS app_user (
crewguard_submission(#     user_id         SERIAL       PRIMARY KEY,
crewguard_submission(#     username        VARCHAR(50)  NOT NULL UNIQUE,
crewguard_submission(#     password_hash   VARCHAR(255) NOT NULL,
crewguard_submission(#     app_role        VARCHAR(20)  NOT NULL,
crewguard_submission(#     created_at      TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
crewguard_submission(#     CONSTRAINT chk_app_role CHECK (app_role IN ('SCHEDULER', 'AUDITOR'))
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS crew_member (
crewguard_submission(#     crew_id         SERIAL       PRIMARY KEY,
crewguard_submission(#     employee_code   VARCHAR(20)  NOT NULL UNIQUE,
crewguard_submission(#     first_name      VARCHAR(60)  NOT NULL,
crewguard_submission(#     last_name       VARCHAR(60)  NOT NULL,
crewguard_submission(#     crew_role       VARCHAR(20)  NOT NULL,
crewguard_submission(#     home_base       CHAR(3)      NOT NULL,
crewguard_submission(#     hire_date       DATE         NOT NULL,
crewguard_submission(#     is_active       BOOLEAN      NOT NULL DEFAULT TRUE,
crewguard_submission(#     CONSTRAINT fk_crew_home_base FOREIGN KEY (home_base) REFERENCES airport(iata_code)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE RESTRICT,
crewguard_submission(#     CONSTRAINT chk_crew_role CHECK (crew_role IN ('CAPTAIN', 'FIRST_OFFICER', 'CABIN_CREW')),
crewguard_submission(#     CONSTRAINT chk_crew_hire_date CHECK (hire_date <= CURRENT_DATE)
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS certification (
crewguard_submission(#     cert_id         SERIAL       PRIMARY KEY,
```

A1 - Extension and first seven tables (incl. exclusion constraint) (contd.)

```
crewguard_submission(#     crew_id         INTEGER      NOT NULL,
crewguard_submission(#     type_code       VARCHAR(10)  NOT NULL,
crewguard_submission(#     issued_on       DATE         NOT NULL,
crewguard_submission(#     expires_on      DATE         NOT NULL,
crewguard_submission(#     CONSTRAINT fk_cert_crew FOREIGN KEY (crew_id) REFERENCES crew_member(crew_id)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE CASCADE,
crewguard_submission(#     CONSTRAINT fk_cert_type FOREIGN KEY (type_code) REFERENCES aircraft_type(type_code)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE RESTRICT,
crewguard_submission(#     CONSTRAINT uk_cert_crew_type_issue UNIQUE (crew_id, type_code, issued_on),
crewguard_submission(#     CONSTRAINT chk_cert_validity_order CHECK (expires_on > issued_on)
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS flight (
crewguard_submission(#     flight_id       SERIAL       PRIMARY KEY,
crewguard_submission(#     flight_number   VARCHAR(10)  NOT NULL,
crewguard_submission(#     type_code       VARCHAR(10)  NOT NULL,
crewguard_submission(#     dep_airport     CHAR(3)      NOT NULL,
crewguard_submission(#     arr_airport     CHAR(3)      NOT NULL,
crewguard_submission(#     sched_dep_utc   TIMESTAMPTZ  NOT NULL,
crewguard_submission(#     sched_arr_utc   TIMESTAMPTZ  NOT NULL,
crewguard_submission(#     CONSTRAINT fk_flight_type FOREIGN KEY (type_code) REFERENCES aircraft_type(type_code)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE RESTRICT,
crewguard_submission(#     CONSTRAINT fk_flight_dep FOREIGN KEY (dep_airport) REFERENCES airport(iata_code)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE RESTRICT,
crewguard_submission(#     CONSTRAINT fk_flight_arr FOREIGN KEY (arr_airport) REFERENCES airport(iata_code)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE RESTRICT,
crewguard_submission(#     CONSTRAINT uk_flight_number_dep UNIQUE (flight_number, sched_dep_utc),
crewguard_submission(#     CONSTRAINT chk_flight_time_order CHECK (sched_arr_utc > sched_dep_utc),
crewguard_submission(#     CONSTRAINT chk_flight_distinct_endpoints CHECK (dep_airport <> arr_airport)
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS duty_assignment (
crewguard_submission(#     assignment_id   SERIAL       PRIMARY KEY,
crewguard_submission(#     crew_id         INTEGER      NOT NULL,
crewguard_submission(#     flight_id       INTEGER      NOT NULL,
crewguard_submission(#     assigned_role   VARCHAR(20)  NOT NULL,
crewguard_submission(#     duty_start_utc  TIMESTAMPTZ  NOT NULL,
crewguard_submission(#     duty_end_utc    TIMESTAMPTZ  NOT NULL,
crewguard_submission(#     created_by      INTEGER,
crewguard_submission(#     created_at      TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
crewguard_submission(#     CONSTRAINT fk_duty_crew FOREIGN KEY (crew_id) REFERENCES crew_member(crew_id)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE CASCADE,
crewguard_submission(#     CONSTRAINT fk_duty_flight FOREIGN KEY (flight_id) REFERENCES flight(flight_id)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE CASCADE,
```

A1 - Extension and first seven tables (incl. exclusion constraint) (contd.)

```
crewguard_submission(#     CONSTRAINT fk_duty_created_by FOREIGN KEY (created_by) REFERENCES app_user(user_id)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE SET NULL,
crewguard_submission(#     CONSTRAINT uk_duty_crew_flight UNIQUE (crew_id, flight_id),
crewguard_submission(#     CONSTRAINT chk_duty_time_order CHECK (duty_end_utc > duty_start_utc),
crewguard_submission(#     CONSTRAINT chk_duty_assigned_role CHECK (assigned_role IN ('CAPTAIN', 'FIRST_OFFICER', 'CABIN_CREW'))
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- A crew member cannot be on two overlapping duties at the same time.
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# ALTER TABLE duty_assignment DROP CONSTRAINT IF EXISTS excl_duty_no_overlap;
NOTICE:  constraint "excl_duty_no_overlap" of relation "duty_assignment" does not exist, skipping
ALTER TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# ALTER TABLE duty_assignment
crewguard_submission-#     ADD CONSTRAINT excl_duty_no_overlap
crewguard_submission-#     EXCLUDE USING GIST (
crewguard_submission(#         crew_id WITH =,
crewguard_submission(#         tstzrange(duty_start_utc, duty_end_utc) WITH &&
crewguard_submission(#     );
ALTER TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

A2 - Rule, validation-run, violation and link tables

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS rule_definition (
crewguard_submission(#     rule_code       CHAR(3)      PRIMARY KEY,
crewguard_submission(#     rule_name       VARCHAR(120) NOT NULL,
crewguard_submission(#     description     TEXT         NOT NULL,
crewguard_submission(#     regulation_ref  VARCHAR(120),
crewguard_submission(#     threshold_value NUMERIC(8,2),
crewguard_submission(#     threshold_unit  VARCHAR(20),
crewguard_submission(#     window_days     INTEGER,
crewguard_submission(#     severity        VARCHAR(10)  NOT NULL,
crewguard_submission(#     is_active       BOOLEAN      NOT NULL DEFAULT TRUE,
crewguard_submission(#     CONSTRAINT chk_rule_severity CHECK (severity IN ('LOW', 'MEDIUM', 'HIGH', 'CRITICAL')),
crewguard_submission(#     CONSTRAINT chk_rule_window_days CHECK (window_days IS NULL OR window_days > 0)
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS validation_run (
crewguard_submission(#     run_id          SERIAL       PRIMARY KEY,
crewguard_submission(#     triggered_by    INTEGER,
crewguard_submission(#     period_start    DATE         NOT NULL,
crewguard_submission(#     period_end      DATE         NOT NULL,
crewguard_submission(#     started_at      TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
crewguard_submission(#     finished_at     TIMESTAMPTZ,
crewguard_submission(#     status          VARCHAR(15)  NOT NULL DEFAULT 'RUNNING',
crewguard_submission(#     violations_found INTEGER     NOT NULL DEFAULT 0,
crewguard_submission(#     CONSTRAINT fk_run_user FOREIGN KEY (triggered_by) REFERENCES app_user(user_id)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE SET NULL,
crewguard_submission(#     CONSTRAINT chk_run_status CHECK (status IN ('RUNNING', 'COMPLETED', 'FAILED', 'STALE')),
crewguard_submission(#     CONSTRAINT chk_run_period_order CHECK (period_end >= period_start),
crewguard_submission(#     CONSTRAINT chk_run_violation_count CHECK (violations_found >= 0)
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS violation (
crewguard_submission(#     violation_id    SERIAL       PRIMARY KEY,
crewguard_submission(#     run_id          INTEGER      NOT NULL,
crewguard_submission(#     rule_code       CHAR(3)      NOT NULL,
crewguard_submission(#     crew_id         INTEGER      NOT NULL,
crewguard_submission(#     severity        VARCHAR(10)  NOT NULL,
crewguard_submission(#     window_start    TIMESTAMPTZ,
crewguard_submission(#     window_end      TIMESTAMPTZ,
crewguard_submission(#     observed_value  NUMERIC(8,2),
crewguard_submission(#     threshold_value NUMERIC(8,2),
crewguard_submission(#     evidence_doc_id VARCHAR(24),
crewguard_submission(#     CONSTRAINT fk_violation_run FOREIGN KEY (run_id) REFERENCES validation_run(run_id)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE CASCADE,
crewguard_submission(#     CONSTRAINT fk_violation_rule FOREIGN KEY (rule_code) REFERENCES rule_definition(rule_code)
```

A2 - Rule, validation-run, violation and link tables (contd.)

```
crewguard_submission(#         ON UPDATE CASCADE ON DELETE RESTRICT,
crewguard_submission(#     CONSTRAINT fk_violation_crew FOREIGN KEY (crew_id) REFERENCES crew_member(crew_id)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE CASCADE,
crewguard_submission(#     CONSTRAINT chk_violation_severity CHECK (severity IN ('LOW', 'MEDIUM', 'HIGH', 'CRITICAL')),
crewguard_submission(#     CONSTRAINT chk_violation_window_order
crewguard_submission(#         CHECK (window_end IS NULL OR window_start IS NULL OR window_end >= window_start)
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TABLE IF NOT EXISTS violation_assignment (
crewguard_submission(#     violation_id       INTEGER   NOT NULL,
crewguard_submission(#     assignment_id      INTEGER   NOT NULL,
crewguard_submission(#     contribution_value NUMERIC(8,2),
crewguard_submission(#     CONSTRAINT pk_violation_assignment PRIMARY KEY (violation_id, assignment_id),
crewguard_submission(#     CONSTRAINT fk_va_violation FOREIGN KEY (violation_id) REFERENCES violation(violation_id)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE CASCADE,
crewguard_submission(#     CONSTRAINT fk_va_assignment FOREIGN KEY (assignment_id) REFERENCES duty_assignment(assignment_id)
crewguard_submission(#         ON UPDATE CASCADE ON DELETE CASCADE
crewguard_submission(# );
CREATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

A3 - List all tables (\dt, commented in source line 209)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- List all tables
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# \dt
              List of relations
 Schema |          Name          | Type  |  Owner
--------+------------------------+-------+----------
 public | aircraft_type          | table | postgres
 public | airport                | table | postgres
 public | app_user               | table | postgres
 public | certification          | table | postgres
 public | crew_member            | table | postgres
 public | duty_assignment        | table | postgres
 public | flight                 | table | postgres
 public | rule_definition        | table | postgres
 public | validation_run         | table | postgres
 public | violation              | table | postgres
 public | violation_assignment   | table | postgres
(11 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART B : SAMPLE DATA (INSERT) -

B1 - TRUNCATE, then airport / aircraft_type / app_user / crew_member

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# TRUNCATE violation_assignment, violation, validation_run, duty_assignment,
crewguard_submission-#          certification, flight, crew_member, app_user, rule_definition,
crewguard_submission-#          aircraft_type, airport RESTART IDENTITY CASCADE;
TRUNCATE TABLE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- airport ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO airport (iata_code, name, city, country, timezone) VALUES
crewguard_submission-# ('BLR', 'Kempegowda International Airport',  'Bengaluru', 'India',               'Asia/Kolkata'),
crewguard_submission-# ('DEL', 'Indira Gandhi International Airport','Delhi',    'India',               'Asia/Kolkata'),
crewguard_submission-# ('BOM', 'Chhatrapati Shivaji Maharaj Intl',  'Mumbai',    'India',               'Asia/Kolkata'),
crewguard_submission-# ('MAA', 'Chennai International Airport',     'Chennai',   'India',               'Asia/Kolkata'),
crewguard_submission-# ('HYD', 'Rajiv Gandhi International Airport','Hyderabad', 'India',               'Asia/Kolkata'),
crewguard_submission-# ('DXB', 'Dubai International Airport',       'Dubai',     'United Arab Emirates', 'Asia/Dubai');
INSERT 0 6
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- aircraft_type ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO aircraft_type (type_code, manufacturer, description) VALUES
crewguard_submission-# ('A320', 'Airbus', 'Airbus A320neo narrow-body'),
crewguard_submission-# ('A321', 'Airbus', 'Airbus A321neo narrow-body'),
crewguard_submission-# ('B738', 'Boeing', 'Boeing 737-800 narrow-body'),
crewguard_submission-# ('B788', 'Boeing', 'Boeing 787-8 wide-body');
INSERT 0 4
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- app_user ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO app_user (user_id, username, password_hash, app_role) VALUES
crewguard_submission-# (1, 'sched_anita',  '$2b$10$QxHlyq0b3eU0mXh5kP2u7e', 'SCHEDULER'),
crewguard_submission-# (2, 'sched_vikram', '$2b$10$Lm9fR2sT4vW6xY8zA0bC1d', 'SCHEDULER'),
crewguard_submission-# (3, 'audit_meera',  '$2b$10$Zn3pQ5rS7tU9vW1xY3zA5b', 'AUDITOR'),
crewguard_submission-# (4, 'audit_rahul',  '$2b$10$Kj2hG4fD6sA8qW0eR2tY4u', 'AUDITOR');
INSERT 0 4
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- crew_member ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO crew_member (crew_id, employee_code, first_name, last_name, crew_role, home_base, hire_date, is_active) VALUES
crewguard_submission-# (1, 'EMP1001', 'Arjun',  'Menon',  'CAPTAIN',       'BLR', '2015-06-01', TRUE),
crewguard_submission-# (2, 'EMP1002', 'Kavya',  'Sharma', 'CAPTAIN',       'BLR', '2017-03-15', TRUE),
crewguard_submission-# (3, 'EMP1003', 'Rohan',  'Kumar',  'FIRST_OFFICER', 'BLR', '2019-07-10', TRUE),
crewguard_submission-# (4, 'EMP1004', 'Sneha',  'Patil',  'FIRST_OFFICER', 'DEL', '2020-01-20', TRUE),
crewguard_submission-# (5, 'EMP1005', 'Vikram', 'Joshi',  'CABIN_CREW',    'BLR', '2021-09-05', TRUE),
crewguard_submission-# (6, 'EMP1006', 'Nisha',  'Shah',   'CABIN_CREW',    'DEL', '2022-02-11', TRUE),
crewguard_submission-# (7, 'EMP1007', 'Aditya', 'Reddy',  'CAPTAIN',       'BOM', '2016-11-30', TRUE),
crewguard_submission-# (8, 'EMP1008', 'Meera',  'Nair',   'CABIN_CREW',    'MAA', '2023-05-18', FALSE);
INSERT 0 8
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

B2 - certification / flight / duty_assignment

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- certification ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO certification (cert_id, crew_id, type_code, issued_on, expires_on) VALUES
crewguard_submission-# (1, 1, 'A320', '2024-01-10', '2027-01-09'),
crewguard_submission-# (2, 1, 'A321', '2024-02-15', '2027-02-14'),
crewguard_submission-# (3, 2, 'A320', '2023-11-01', '2026-10-31'),
crewguard_submission-# (4, 2, 'B738', '2025-01-05', '2028-01-04'),
crewguard_submission-# (5, 3, 'A320', '2024-05-20', '2027-05-19'),
crewguard_submission-# (6, 3, 'A321', '2025-06-18', '2028-06-17'),
crewguard_submission-# (7, 4, 'B738', '2023-08-12', '2026-08-11'),   -- expired certificate
crewguard_submission-# (8, 7, 'B788', '2024-03-01', '2027-02-28');
INSERT 0 8
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- flight ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO flight (flight_id, flight_number, type_code, dep_airport, arr_airport, sched_dep_utc, sched_arr_utc) VALUES
crewguard_submission-# (1,  'CG101', 'A320', 'BLR', 'DEL', '2026-09-01 01:00+00', '2026-09-01 03:45+00'),
crewguard_submission-# (2,  'CG102', 'A320', 'DEL', 'BLR', '2026-09-01 05:30+00', '2026-09-01 08:15+00'),
crewguard_submission-# (3,  'CG201', 'B738', 'BLR', 'BOM', '2026-09-02 02:00+00', '2026-09-02 03:45+00'),
crewguard_submission-# (4,  'CG202', 'B738', 'BOM', 'BLR', '2026-09-02 05:00+00', '2026-09-02 06:45+00'),
crewguard_submission-# (5,  'CG301', 'A321', 'BLR', 'MAA', '2026-09-03 01:30+00', '2026-09-03 02:40+00'),
crewguard_submission-# (6,  'CG302', 'A321', 'MAA', 'BLR', '2026-09-03 04:00+00', '2026-09-03 05:10+00'),
crewguard_submission-# (7,  'CG401', 'B788', 'BOM', 'DXB', '2026-09-04 18:00+00', '2026-09-04 21:30+00'),
crewguard_submission-# (8,  'CG402', 'B788', 'DXB', 'BOM', '2026-09-05 01:00+00', '2026-09-05 04:30+00'),
crewguard_submission-# (9,  'CG501', 'A320', 'BLR', 'HYD', '2026-09-06 02:00+00', '2026-09-06 03:00+00'),
crewguard_submission-# (10, 'CG502', 'A320', 'HYD', 'BLR', '2026-09-06 04:30+00', '2026-09-06 05:30+00');
INSERT 0 10
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- duty_assignment ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- Duty starts 1 h before departure and ends 30 min after arrival.
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO duty_assignment (assignment_id, crew_id, flight_id, assigned_role, duty_start_utc, duty_end_utc, created_by) VALUES
crewguard_submission-# (1,  1, 1,  'CAPTAIN',       '2026-09-01 00:00+00', '2026-09-01 04:15+00', 1),
crewguard_submission-# (2,  1, 2,  'CAPTAIN',       '2026-09-01 04:30+00', '2026-09-01 08:45+00', 1),
crewguard_submission-# (3,  1, 5,  'CAPTAIN',       '2026-09-03 00:30+00', '2026-09-03 03:10+00', 1),
crewguard_submission-# (4,  1, 6,  'CAPTAIN',       '2026-09-03 03:15+00', '2026-09-03 05:40+00', 1),
crewguard_submission-# (5,  2, 3,  'CAPTAIN',       '2026-09-02 01:00+00', '2026-09-02 04:15+00', 1),
crewguard_submission-# (6,  2, 4,  'CAPTAIN',       '2026-09-02 04:20+00', '2026-09-02 07:15+00', 1),
crewguard_submission-# (7,  2, 9,  'CAPTAIN',       '2026-09-06 01:00+00', '2026-09-06 03:30+00', 2),
crewguard_submission-# (8,  2, 10, 'CAPTAIN',       '2026-09-06 03:35+00', '2026-09-06 06:00+00', 2),
crewguard_submission-# (9,  3, 1,  'FIRST_OFFICER', '2026-09-01 00:00+00', '2026-09-01 04:15+00', 1),
crewguard_submission-# (10, 3, 2,  'FIRST_OFFICER', '2026-09-01 04:30+00', '2026-09-01 08:45+00', 1),
crewguard_submission-# (11, 3, 3,  'FIRST_OFFICER', '2026-09-02 01:00+00', '2026-09-02 04:15+00', 1),
crewguard_submission-# (12, 3, 4,  'FIRST_OFFICER', '2026-09-02 04:20+00', '2026-09-02 07:15+00', 1),
crewguard_submission-# (13, 4, 5,  'FIRST_OFFICER', '2026-09-03 00:30+00', '2026-09-03 03:10+00', 2),
crewguard_submission-# (14, 4, 6,  'FIRST_OFFICER', '2026-09-03 03:15+00', '2026-09-03 05:40+00', 2),
crewguard_submission-# (15, 5, 1,  'CABIN_CREW',    '2026-09-01 00:00+00', '2026-09-01 04:15+00', 1),
crewguard_submission-# (16, 5, 2,  'CABIN_CREW',    '2026-09-01 04:30+00', '2026-09-01 08:45+00', 1),
crewguard_submission-# (17, 5, 9,  'CABIN_CREW',    '2026-09-06 01:00+00', '2026-09-06 03:30+00', 2),
```

B2 - certification / flight / duty_assignment (contd.)

```
crewguard_submission-# (18, 5, 10, 'CABIN_CREW',    '2026-09-06 03:35+00', '2026-09-06 06:00+00', 2),
crewguard_submission-# (19, 6, 3,  'CABIN_CREW',    '2026-09-02 01:00+00', '2026-09-02 04:15+00', 1),
crewguard_submission-# (20, 6, 4,  'CABIN_CREW',    '2026-09-02 04:20+00', '2026-09-02 07:15+00', 1),
crewguard_submission-# (21, 7, 7,  'CAPTAIN',       '2026-09-04 17:00+00', '2026-09-04 22:00+00', 2),
crewguard_submission-# (22, 7, 8,  'CAPTAIN',       '2026-09-05 00:00+00', '2026-09-05 05:00+00', 2);
INSERT 0 22
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

B3 - rule_definition / validation_run / violation / violation_assignment

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- rule_definition ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO rule_definition (rule_code, rule_name, description, regulation_ref, threshold_value, threshold_unit, window_days, severity, is_active) VALUES
crewguard_submission-# ('RST', 'Minimum Rest Between Duties',
crewguard_submission(#         'A crew member must receive a minimum continuous rest period between the end of one duty and the start of the next duty.',
crewguard_submission(#         'DGCA CAR Section 7 Series J Part III', 10.00, 'HOURS', 1, 'HIGH', TRUE),
crewguard_submission-# ('FDP', 'Maximum Flight Duty Period',
crewguard_submission(#         'The flight duty period of a single duty must not exceed the maximum permitted duration for the crew composition.',
crewguard_submission(#         'DGCA CAR Section 7 Series J Part III', 13.00, 'HOURS', 1, 'CRITICAL', TRUE),
crewguard_submission-# ('WDH', 'Maximum Duty Hours in Seven Days',
crewguard_submission(#         'Cumulative duty hours in any rolling window of seven consecutive days must not exceed the prescribed limit.',
crewguard_submission(#         'DGCA CAR Section 7 Series J Part IV', 60.00, 'HOURS', 7, 'MEDIUM', TRUE),
crewguard_submission-# ('MDH', 'Maximum Duty Hours in Twenty Eight Days',
crewguard_submission(#         'Cumulative duty hours in any rolling window of twenty eight consecutive days must not exceed the prescribed limit.',
crewguard_submission(#         'DGCA CAR Section 7 Series J Part IV', 190.00, 'HOURS', 28, 'MEDIUM', TRUE),
crewguard_submission-# ('CER', 'Valid Type Certification Required',
crewguard_submission(#         'A crew member must hold a valid and unexpired type certification for the aircraft type operated on the duty.',
crewguard_submission(#         'DGCA CAR Section 2 Series X Part I', NULL, NULL, NULL, 'CRITICAL', TRUE),
crewguard_submission-# ('WOF', 'Weekly Day Off Required',
crewguard_submission(#         'A crew member must be rostered at least one full day free of all duty in every seven day period.',
crewguard_submission(#         'DGCA CAR Section 7 Series J Part IV', 1.00, 'DAYS', 7, 'LOW', FALSE);
INSERT 0 6
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- validation_run ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO validation_run (run_id, triggered_by, period_start, period_end, started_at, finished_at, status, violations_found) VALUES
crewguard_submission-# (1, 3, '2026-09-01', '2026-09-07', '2026-09-08 04:00+00', '2026-09-08 04:02+00', 'COMPLETED', 3),
crewguard_submission-# (2, 4, '2026-09-08', '2026-09-14', '2026-09-15 04:00+00', '2026-09-15 04:01+00', 'COMPLETED', 0),
crewguard_submission-# (3, 3, '2026-09-15', '2026-09-21', '2026-09-22 04:00+00', NULL,                  'RUNNING',   0);
INSERT 0 3
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- violation ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO violation (violation_id, run_id, rule_code, crew_id, severity, window_start, window_end, observed_value, threshold_value, evidence_doc_id) VALUES
crewguard_submission-# (1, 1, 'RST', 1, 'HIGH',     '2026-09-01 04:15+00', '2026-09-01 04:30+00',  0.25, 10.00, 'DOC-2026-0001'),
crewguard_submission-# (2, 1, 'RST', 3, 'HIGH',     '2026-09-01 04:15+00', '2026-09-01 04:30+00',  0.25, 10.00, 'DOC-2026-0002'),
crewguard_submission-# (3, 1, 'CER', 4, 'CRITICAL', '2026-09-03 00:30+00', '2026-09-03 05:40+00',  NULL,  NULL, 'DOC-2026-0003');
INSERT 0 3
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- violation_assignment ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO violation_assignment (violation_id, assignment_id, contribution_value) VALUES
crewguard_submission-# (1, 1,  4.25),
crewguard_submission-# (1, 2,  4.25),
crewguard_submission-# (2, 9,  4.25),
crewguard_submission-# (2, 10, 4.25),
crewguard_submission-# (3, 13, 2.67),
crewguard_submission-# (3, 14, 2.42);
INSERT 0 6
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

B4 - Re-synchronise sequences

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- ---------- re-synchronise sequences ----------
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT setval('app_user_user_id_seq',             (SELECT MAX(user_id)       FROM app_user));
 setval
--------
      4
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT setval('crew_member_crew_id_seq',          (SELECT MAX(crew_id)       FROM crew_member));
 setval
--------
      8
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT setval('certification_cert_id_seq',        (SELECT MAX(cert_id)       FROM certification));
 setval
--------
      8
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT setval('flight_flight_id_seq',             (SELECT MAX(flight_id)     FROM flight));
 setval
--------
     10
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT setval('duty_assignment_assignment_id_seq',(SELECT MAX(assignment_id) FROM duty_assignment));
 setval
--------
     22
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT setval('validation_run_run_id_seq',        (SELECT MAX(run_id)        FROM validation_run));
 setval
--------
      3
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT setval('violation_violation_id_seq',       (SELECT MAX(violation_id)  FROM violation));
 setval
--------
      3
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

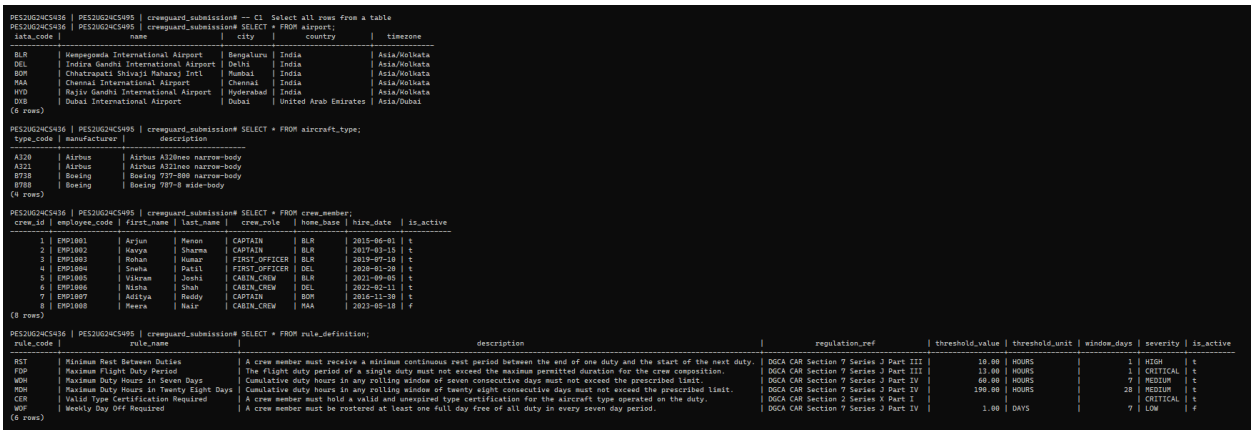
B5 - Row counts after loading

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- Row counts after loading
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT 'airport' AS table_name, COUNT(*) FROM airport
crewguard_submission-# UNION ALL SELECT 'aircraft_type',        COUNT(*) FROM aircraft_type
crewguard_submission-# UNION ALL SELECT 'app_user',             COUNT(*) FROM app_user
crewguard_submission-# UNION ALL SELECT 'crew_member',          COUNT(*) FROM crew_member
crewguard_submission-# UNION ALL SELECT 'certification',        COUNT(*) FROM certification
crewguard_submission-# UNION ALL SELECT 'flight',               COUNT(*) FROM flight
crewguard_submission-# UNION ALL SELECT 'duty_assignment',      COUNT(*) FROM duty_assignment
crewguard_submission-# UNION ALL SELECT 'rule_definition',      COUNT(*) FROM rule_definition
crewguard_submission-# UNION ALL SELECT 'validation_run',       COUNT(*) FROM validation_run
crewguard_submission-# UNION ALL SELECT 'violation',            COUNT(*) FROM violation
crewguard_submission-# UNION ALL SELECT 'violation_assignment', COUNT(*) FROM violation_assignment
crewguard_submission-# ORDER BY 1;
      table_name       | count
-----------------------+-------
 aircraft_type         |     4
 airport               |     6
 app_user              |     4
 certification         |     8
 crew_member           |     8
 duty_assignment       |    22
 flight                |    10
 rule_definition       |     6
 validation_run        |     3
 violation             |     3
 violation_assignment  |     6
(11 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART C : BASIC RETRIEVAL -

C1 - Select all rows from a table



```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C1  Select all rows from a table
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT * FROM airport;
 iata_code |                 name                 |   city    |       country        |   timezone
-----------+--------------------------------------+-----------+----------------------+---------------
 BLR       | Kempegowda International Airport     | Bengaluru | India                | Asia/Kolkata
 DEL       | Indira Gandhi International Airport  | Delhi     | India                | Asia/Kolkata
 BOM       | Chhatrapati Shivaji Maharaj Intl     | Mumbai    | India                | Asia/Kolkata
 MAA       | Chennai International Airport        | Chennai   | India                | Asia/Kolkata
 HYD       | Rajiv Gandhi International Airport   | Hyderabad | India                | Asia/Kolkata
 DXB       | Dubai International Airport          | Dubai     | United Arab Emirates | Asia/Dubai
(6 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT * FROM aircraft_type;
 type_code | manufacturer |          description
-----------+--------------+-------------------------
 A320      | Airbus       | Airbus A320neo narrow-body
 A321      | Airbus       | Airbus A321neo narrow-body
 B738      | Boeing       | Boeing 737-800 narrow-body
 B788      | Boeing       | Boeing 787-8 wide-body
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT * FROM crew_member;
 crew_id | employee_code | first_name | last_name |  crew_role   | home_base | hire_date  | is_active
---------+---------------+------------+-----------+--------------+-----------+------------+-----------
       1 | EMP1001       | Arjun      | Menon     | CAPTAIN      | BLR       | 2015-06-01 | t
       2 | EMP1002       | Kavya      | Sharma    | CAPTAIN      | BLR       | 2017-03-15 | t
       3 | EMP1003       | Rohan      | Kumar     | FIRST_OFFICER | BLR       | 2019-07-10 | t
       4 | EMP1004       | Sneha      | Patil     | FIRST_OFFICER | DEL       | 2020-01-20 | t
       5 | EMP1005       | Vikram     | Joshi     | CABIN_CREW   | BLR       | 2021-09-05 | t
       6 | EMP1006       | Nisha      | Shah      | CABIN_CREW   | DEL       | 2022-02-11 | t
       7 | EMP1007       | Aditya     | Reddy     | CAPTAIN      | BOM       | 2016-11-30 | t
       8 | EMP1008       | Meera      | Nair      | CABIN_CREW   | MAA       | 2023-05-18 | f
(8 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT * FROM rule_definition;
 rule_code |             rule_name              |                                                  description                                                  |          regulation_ref          | threshold_value | threshold_unit | window_days | severity | is_active
-----------+------------------------------------+---------------------------------------------------------------------------------------------------------------+----------------------------------+-----------------+----------------+-------------+----------+-----------
 RST       | Minimum Rest Between Duties        | A crew member must receive a minimum continuous rest period between the end of one duty and the start of the next duty. | DGCA CAR Section 7 Series J Part III |           10.00 | HOURS          |           1 | HIGH     | t
 FDP       | Maximum Flight Duty Period         | The flight duty period of a single duty must not exceed the maximum permitted duration for the crew composition.       | DGCA CAR Section 7 Series J Part III |           13.00 | HOURS          |           1 | CRITICAL | t
 WDH       | Maximum Duty Hours in Seven Days   | Cumulative duty hours in any rolling window of seven consecutive days must not exceed the prescribed limit.           | DGCA CAR Section 7 Series J Part IV  |           60.00 | HOURS          |           7 | MEDIUM   | t
 MDH       | Maximum Duty Hours in Twenty Eight Days | Cumulative duty hours in any rolling window of twenty eight consecutive days must not exceed the prescribed limit. | DGCA CAR Section 7 Series J Part IV  |          190.00 | HOURS          |          28 | MEDIUM   | t
 CER       | Valid Type Certification Required  | A crew member must hold a valid and unexpired type certification for the aircraft type operated on the duty.        | DGCA CAR Section 2 Series X Part I   |                 |                |             | CRITICAL | t
 WOF       | Weekly Day Off Required            | A crew member must be rostered at least one full day free of all duty in every seven day period.                    | DGCA CAR Section 7 Series J Part IV  |            1.00 | DAYS           |           7 | LOW      | f
(6 rows)
```

C2 - Projection of selected columns

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C2  Projection of selected columns
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT crew_id, employee_code, first_name, last_name, crew_role, home_base
crewguard_submission-# FROM crew_member;
 crew_id | employee_code | first_name | last_name |   crew_role   | home_base
---------+---------------+------------+-----------+---------------+-----------
       1 | EMP1001       | Arjun      | Menon     | CAPTAIN       | BLR
       2 | EMP1002       | Kavya      | Sharma    | CAPTAIN       | BLR
       3 | EMP1003       | Rohan      | Kumar     | FIRST_OFFICER | BLR
       4 | EMP1004       | Sneha      | Patil     | FIRST_OFFICER | DEL
       5 | EMP1005       | Vikram     | Joshi     | CABIN_CREW    | BLR
       6 | EMP1006       | Nisha      | Shah      | CABIN_CREW    | DEL
       7 | EMP1007       | Aditya     | Reddy     | CAPTAIN       | BOM
       8 | EMP1008       | Meera      | Nair      | CABIN_CREW    | MAA
(8 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

C3 - WHERE with equality

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C3  WHERE with equality
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT crew_id, first_name, last_name, home_base
crewguard_submission-# FROM crew_member
crewguard_submission-# WHERE crew_role = 'CAPTAIN';
 crew_id | first_name | last_name | home_base
---------+------------+-----------+-----------
       1 | Arjun      | Menon     | BLR
       2 | Kavya      | Sharma    | BLR
       7 | Aditya     | Reddy     | BOM
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

C4 - WHERE with a boolean column

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C4  WHERE with a boolean column
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT employee_code, first_name, last_name
crewguard_submission-# FROM crew_member
crewguard_submission-# WHERE is_active = FALSE;
 employee_code | first_name | last_name
---------------+------------+-----------
 EMP1008       | Meera      | Nair
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

C5 - DISTINCT

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C5  DISTINCT
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT DISTINCT crew_role FROM crew_member;
   crew_role
---------------
 CABIN_CREW
 FIRST_OFFICER
 CAPTAIN
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT DISTINCT type_code FROM flight;
 type_code
-----------
 A321
 B788
 B738
 A320
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

C6 - ORDER BY

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C6  ORDER BY
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT employee_code, first_name, last_name, hire_date
crewguard_submission-# FROM crew_member
crewguard_submission-# ORDER BY hire_date ASC;
 employee_code | first_name | last_name | hire_date
---------------+------------+-----------+------------
 EMP1001       | Arjun      | Menon     | 2015-06-01
 EMP1007       | Aditya     | Reddy     | 2016-11-30
 EMP1002       | Kavya      | Sharma    | 2017-03-15
 EMP1003       | Rohan      | Kumar     | 2019-07-10
 EMP1004       | Sneha      | Patil     | 2020-01-20
 EMP1005       | Vikram     | Joshi     | 2021-09-05
 EMP1006       | Nisha      | Shah      | 2022-02-11
 EMP1008       | Meera      | Nair      | 2023-05-18
(8 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT flight_number, dep_airport, arr_airport, sched_dep_utc
crewguard_submission-# FROM flight
crewguard_submission-# ORDER BY sched_dep_utc DESC;
 flight_number | dep_airport | arr_airport |       sched_dep_utc
---------------+-------------+-------------+---------------------------
 CG502         | HYD         | BLR         | 2026-09-06 10:00:00+05:30
 CG501         | BLR         | HYD         | 2026-09-06 07:30:00+05:30
 CG402         | DXB         | BOM         | 2026-09-05 06:30:00+05:30
 CG401         | BOM         | DXB         | 2026-09-04 23:30:00+05:30
 CG302         | MAA         | BLR         | 2026-09-03 09:30:00+05:30
 CG301         | BLR         | MAA         | 2026-09-03 07:00:00+05:30
 CG202         | BOM         | BLR         | 2026-09-02 10:30:00+05:30
 CG201         | BLR         | BOM         | 2026-09-02 07:30:00+05:30
 CG102         | DEL         | BLR         | 2026-09-01 11:00:00+05:30
 CG101         | BLR         | DEL         | 2026-09-01 06:30:00+05:30
(10 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

C7 - LIKE / ILIKE pattern matching

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C7  LIKE / ILIKE pattern matching
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT iata_code, name, city
crewguard_submission-# FROM airport
crewguard_submission-# WHERE name LIKE '%International%';
 iata_code |               name                |   city
-----------+-----------------------------------+-----------
 BLR       | Kempegowda International Airport    | Bengaluru
 DEL       | Indira Gandhi International Airport | Delhi
 MAA       | Chennai International Airport       | Chennai
 HYD       | Rajiv Gandhi International Airport  | Hyderabad
 DXB       | Dubai International Airport         | Dubai
(5 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT rule_code, rule_name
crewguard_submission-# FROM rule_definition
crewguard_submission-# WHERE rule_name ILIKE '%duty%';
 rule_code |               rule_name
-----------+-----------------------------------------
 FDP       | Maximum Flight Duty Period
 WDH       | Maximum Duty Hours in Seven Days
 MDH       | Maximum Duty Hours in Twenty Eight Days
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

C8 - IN

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C8  IN
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT flight_id, flight_number, dep_airport, arr_airport
crewguard_submission-# FROM flight
crewguard_submission-# WHERE dep_airport IN ('BLR', 'BOM');
 flight_id | flight_number | dep_airport | arr_airport
-----------+---------------+-------------+-------------
         1 | CG101         | BLR         | DEL
         3 | CG201         | BLR         | BOM
         4 | CG202         | BOM         | BLR
         5 | CG301         | BLR         | MAA
         7 | CG401         | BOM         | DXB
         9 | CG501         | BLR         | HYD
(6 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT rule_code, rule_name, severity
crewguard_submission-# FROM rule_definition
crewguard_submission-# WHERE severity IN ('HIGH', 'CRITICAL');
 rule_code |             rule_name             | severity
-----------+-----------------------------------+----------
 RST       | Minimum Rest Between Duties       | HIGH
 FDP       | Maximum Flight Duty Period        | CRITICAL
 CER       | Valid Type Certification Required | CRITICAL
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

C9 - BETWEEN

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C9  BETWEEN
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT flight_number, sched_dep_utc, sched_arr_utc
crewguard_submission-# FROM flight
crewguard_submission-# WHERE sched_dep_utc BETWEEN '2026-09-01' AND '2026-09-04';
 flight_number |       sched_dep_utc       |       sched_arr_utc
---------------+---------------------------+---------------------------
 CG101         | 2026-09-01 06:30:00+05:30 | 2026-09-01 09:15:00+05:30
 CG102         | 2026-09-01 11:00:00+05:30 | 2026-09-01 13:45:00+05:30
 CG201         | 2026-09-02 07:30:00+05:30 | 2026-09-02 09:15:00+05:30
 CG202         | 2026-09-02 10:30:00+05:30 | 2026-09-02 12:15:00+05:30
 CG301         | 2026-09-03 07:00:00+05:30 | 2026-09-03 08:10:00+05:30
 CG302         | 2026-09-03 09:30:00+05:30 | 2026-09-03 10:40:00+05:30
(6 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT cert_id, crew_id, type_code, expires_on
crewguard_submission-# FROM certification
crewguard_submission-# WHERE expires_on BETWEEN '2026-01-01' AND '2026-12-31';
 cert_id | crew_id | type_code | expires_on
---------+---------+-----------+------------
       3 |       2 | A320      | 2026-10-31
       7 |       4 | B738      | 2026-08-11
(2 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

C10 - IS NULL / IS NOT NULL

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C10 IS NULL / IS NOT NULL
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT run_id, period_start, period_end, status, finished_at
crewguard_submission-# FROM validation_run
crewguard_submission-# WHERE finished_at IS NULL;
 run_id | period_start | period_end | status  | finished_at
--------+--------------+------------+---------+-------------
      3 | 2026-09-15   | 2026-09-21 | RUNNING |
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT violation_id, rule_code, observed_value
crewguard_submission-# FROM violation
crewguard_submission-# WHERE observed_value IS NOT NULL;
 violation_id | rule_code | observed_value
--------------+-----------+----------------
            1 | RST       |           0.25
            2 | RST       |           0.25
(2 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

C11 - Compound conditions, computed column and LIMIT

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- C11 Compound conditions, computed column and LIMIT
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT flight_number,
crewguard_submission-#        dep_airport || ' -> ' || arr_airport                    AS sector,
crewguard_submission-#        sched_arr_utc - sched_dep_utc                          AS block_time
crewguard_submission-# FROM flight
crewguard_submission-# WHERE type_code = 'A320' AND dep_airport <> 'DEL'
crewguard_submission-# ORDER BY sched_dep_utc
crewguard_submission-# LIMIT 5;
 flight_number |   sector   | block_time
---------------+------------+------------
 CG101         | BLR -> DEL | 02:45:00
 CG501         | BLR -> HYD | 01:00:00
 CG502         | HYD -> BLR | 01:00:00
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART D : UPDATE AND DELETE -

D1 - UPDATE a single row (RETURNING shows the changed row)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- D1  UPDATE a single row (RETURNING shows the changed row)
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# UPDATE crew_member
crewguard_submission-# SET is_active = TRUE
crewguard_submission-# WHERE crew_id = 8
crewguard_submission-# RETURNING crew_id, employee_code, is_active;
 crew_id | employee_code | is_active
---------+---------------+-----------
       8 | EMP1008       | t
(1 row)

UPDATE 1
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

D2 - UPDATE using an expression

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- D2  UPDATE using an expression
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# UPDATE flight
crewguard_submission-# SET sched_arr_utc = sched_arr_utc + INTERVAL '20 minutes'
crewguard_submission-# WHERE flight_id = 9
crewguard_submission-# RETURNING flight_id, flight_number, sched_dep_utc, sched_arr_utc;
 flight_id | flight_number |       sched_dep_utc       |       sched_arr_utc
-----------+---------------+---------------------------+---------------------------
         9 | CG501         | 2026-09-06 07:30:00+05:30 | 2026-09-06 08:50:00+05:30
(1 row)

UPDATE 1
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

D3 - UPDATE several rows at once

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- D3  UPDATE several rows at once
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# UPDATE rule_definition
crewguard_submission-# SET is_active = TRUE
crewguard_submission-# WHERE rule_code = 'WOF'
crewguard_submission-# RETURNING rule_code, rule_name, is_active;
 rule_code |        rule_name        | is_active
-----------+-------------------------+-----------
 WOF       | Weekly Day Off Required | t
(1 row)

UPDATE 1
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

D4 - UPDATE driven by a sub-query

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- D4  UPDATE driven by a sub-query
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# UPDATE validation_run
crewguard_submission-# SET status = 'STALE'
crewguard_submission-# WHERE run_id IN (SELECT run_id FROM validation_run
crewguard_submission(#                 WHERE status = 'RUNNING' AND started_at < NOW() - INTERVAL '1 day')
crewguard_submission-# RETURNING run_id, status;
 run_id | status
--------+--------
      3 | STALE
(1 row)

UPDATE 1
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

D5 - DELETE a single row

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- D5  DELETE a single row
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# DELETE FROM certification
crewguard_submission-# WHERE cert_id = 6
crewguard_submission-# RETURNING cert_id, crew_id, type_code;
 cert_id | crew_id | type_code
---------+---------+-----------
       6 |       3 | A321
(1 row)

DELETE 1
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

D6 - Verify the effect of the updates

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- D6  Verify the effect of the updates
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT crew_id, employee_code, is_active FROM crew_member WHERE crew_id = 8;
 crew_id | employee_code | is_active
---------+---------------+-----------
       8 | EMP1008       | t
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT rule_code, is_active FROM rule_definition ORDER BY rule_code;
 rule_code | is_active
-----------+-----------
 CER       | t
 FDP       | t
 MDH       | t
 RST       | t
 WDH       | t
 WOF       | t
(6 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT run_id, status, finished_at FROM validation_run ORDER BY run_id;
 run_id |  status   |        finished_at
--------+-----------+---------------------------
      1 | COMPLETED | 2026-09-08 09:32:00+05:30
      2 | COMPLETED | 2026-09-15 09:31:00+05:30
      3 | STALE     |
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART E : JOINS -

E1 - INNER JOIN - crew duty roster

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- E1  INNER JOIN - crew duty roster
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT da.assignment_id,
crewguard_submission-#        cm.employee_code,
crewguard_submission-#        cm.first_name || ' ' || cm.last_name AS crew_name,
crewguard_submission-#        da.assigned_role,
crewguard_submission-#        f.flight_number,
crewguard_submission-#        f.dep_airport,
crewguard_submission-#        f.arr_airport,
crewguard_submission-#        da.duty_start_utc,
crewguard_submission-#        da.duty_end_utc
crewguard_submission-# FROM duty_assignment da
crewguard_submission-# JOIN crew_member cm ON da.crew_id   = cm.crew_id
crewguard_submission-# JOIN flight      f  ON da.flight_id = f.flight_id
crewguard_submission-# ORDER BY da.duty_start_utc, cm.employee_code;
 assignment_id | employee_code |  crew_name   | assigned_role | flight_number | dep_airport | arr_airport |      duty_start_utc       |       duty_end_utc
---------------+---------------+--------------+---------------+---------------+-------------+-------------+---------------------------+---------------------------
             1 | EMP1001       | Arjun Menon  | CAPTAIN       | CG101         | BLR         | DEL         | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30
             9 | EMP1003       | Rohan Kumar  | FIRST_OFFICER | CG101         | BLR         | DEL         | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30
            15 | EMP1005       | Vikram Joshi | CABIN_CREW    | CG101         | BLR         | DEL         | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30
             2 | EMP1001       | Arjun Menon  | CAPTAIN       | CG102         | DEL         | BLR         | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30
            10 | EMP1003       | Rohan Kumar  | FIRST_OFFICER | CG102         | DEL         | BLR         | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30
            16 | EMP1005       | Vikram Joshi | CABIN_CREW    | CG102         | DEL         | BLR         | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30
             5 | EMP1002       | Kavya Sharma | CAPTAIN       | CG201         | BLR         | BOM         | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30
            11 | EMP1003       | Rohan Kumar  | FIRST_OFFICER | CG201         | BLR         | BOM         | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30
            19 | EMP1006       | Nisha Shah   | CABIN_CREW    | CG201         | BLR         | BOM         | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30
             6 | EMP1002       | Kavya Sharma | CAPTAIN       | CG202         | BOM         | BLR         | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30
            12 | EMP1003       | Rohan Kumar  | FIRST_OFFICER | CG202         | BOM         | BLR         | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30
            20 | EMP1006       | Nisha Shah   | CABIN_CREW    | CG202         | BOM         | BLR         | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30
             3 | EMP1001       | Arjun Menon  | CAPTAIN       | CG301         | BLR         | MAA         | 2026-09-03 06:00:00+05:30 | 2026-09-03 08:40:00+05:30
            13 | EMP1004       | Sneha Patil  | FIRST_OFFICER | CG301         | BLR         | MAA         | 2026-09-03 06:00:00+05:30 | 2026-09-03 08:40:00+05:30
             4 | EMP1001       | Arjun Menon  | CAPTAIN       | CG302         | MAA         | BLR         | 2026-09-03 08:45:00+05:30 | 2026-09-03 11:10:00+05:30
            14 | EMP1004       | Sneha Patil  | FIRST_OFFICER | CG302         | MAA         | BLR         | 2026-09-03 08:45:00+05:30 | 2026-09-03 11:10:00+05:30
            21 | EMP1007       | Aditya Reddy | CAPTAIN       | CG401         | BOM         | DXB         | 2026-09-04 22:30:00+05:30 | 2026-09-05 03:30:00+05:30
            22 | EMP1007       | Aditya Reddy | CAPTAIN       | CG402         | DXB         | BOM         | 2026-09-05 05:30:00+05:30 | 2026-09-05 10:30:00+05:30
             7 | EMP1002       | Kavya Sharma | CAPTAIN       | CG501         | BLR         | HYD         | 2026-09-06 06:30:00+05:30 | 2026-09-06 09:00:00+05:30
            17 | EMP1005       | Vikram Joshi | CABIN_CREW    | CG501         | BLR         | HYD         | 2026-09-06 06:30:00+05:30 | 2026-09-06 09:00:00+05:30
             8 | EMP1002       | Kavya Sharma | CAPTAIN       | CG502         | HYD         | BLR         | 2026-09-06 09:05:00+05:30 | 2026-09-06 11:30:00+05:30
            18 | EMP1005       | Vikram Joshi | CABIN_CREW    | CG502         | HYD         | BLR         | 2026-09-06 09:05:00+05:30 | 2026-09-06 11:30:00+05:30
(22 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

E2 - Five table join - full operational picture of each duty

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- E2  Five table join - full operational picture of each duty
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT f.flight_number,
crewguard_submission-#        dep.city  AS departure_city,
crewguard_submission-#        arr.city  AS arrival_city,
crewguard_submission-#        at.manufacturer || ' ' || at.type_code AS aircraft,
crewguard_submission-#        cm.first_name || ' ' || cm.last_name   AS crew_name,
crewguard_submission-#        da.assigned_role
crewguard_submission-# FROM duty_assignment da
crewguard_submission-# JOIN flight        f   ON da.flight_id  = f.flight_id
crewguard_submission-# JOIN airport       dep ON f.dep_airport = dep.iata_code
crewguard_submission-# JOIN airport       arr ON f.arr_airport = arr.iata_code
crewguard_submission-# JOIN aircraft_type at  ON f.type_code   = at.type_code
crewguard_submission-# JOIN crew_member   cm  ON da.crew_id    = cm.crew_id
crewguard_submission-# ORDER BY f.sched_dep_utc, cm.employee_code;
 flight_number | departure_city | arrival_city |  aircraft   |  crew_name   | assigned_role
---------------+----------------+--------------+-------------+--------------+---------------
 CG101         | Bengaluru      | Delhi        | Airbus A320 | Arjun Menon  | CAPTAIN
 CG101         | Bengaluru      | Delhi        | Airbus A320 | Rohan Kumar  | FIRST_OFFICER
 CG101         | Bengaluru      | Delhi        | Airbus A320 | Vikram Joshi | CABIN_CREW
 CG102         | Delhi          | Bengaluru    | Airbus A320 | Arjun Menon  | CAPTAIN
 CG102         | Delhi          | Bengaluru    | Airbus A320 | Rohan Kumar  | FIRST_OFFICER
 CG102         | Delhi          | Bengaluru    | Airbus A320 | Vikram Joshi | CABIN_CREW
 CG201         | Bengaluru      | Mumbai       | Boeing B738 | Kavya Sharma | CAPTAIN
 CG201         | Bengaluru      | Mumbai       | Boeing B738 | Rohan Kumar  | FIRST_OFFICER
 CG201         | Bengaluru      | Mumbai       | Boeing B738 | Nisha Shah   | CABIN_CREW
 CG202         | Mumbai         | Bengaluru    | Boeing B738 | Kavya Sharma | CAPTAIN
 CG202         | Mumbai         | Bengaluru    | Boeing B738 | Rohan Kumar  | FIRST_OFFICER
 CG202         | Mumbai         | Bengaluru    | Boeing B738 | Nisha Shah   | CABIN_CREW
 CG301         | Bengaluru      | Chennai      | Airbus A321 | Arjun Menon  | CAPTAIN
 CG301         | Bengaluru      | Chennai      | Airbus A321 | Sneha Patil  | FIRST_OFFICER
 CG302         | Chennai        | Bengaluru    | Airbus A321 | Arjun Menon  | CAPTAIN
 CG302         | Chennai        | Bengaluru    | Airbus A321 | Sneha Patil  | FIRST_OFFICER
 CG401         | Mumbai         | Dubai        | Boeing B788 | Aditya Reddy | CAPTAIN
 CG402         | Dubai          | Mumbai       | Boeing B788 | Aditya Reddy | CAPTAIN
 CG501         | Bengaluru      | Hyderabad    | Airbus A320 | Kavya Sharma | CAPTAIN
 CG501         | Bengaluru      | Hyderabad    | Airbus A320 | Vikram Joshi | CABIN_CREW
 CG502         | Hyderabad      | Bengaluru    | Airbus A320 | Kavya Sharma | CAPTAIN
 CG502         | Hyderabad      | Bengaluru    | Airbus A320 | Vikram Joshi | CABIN_CREW
(22 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

E3 - LEFT JOIN - every crew member, including those with no duty

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- E3  LEFT JOIN - every crew member, including those with no duty
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT cm.employee_code,
crewguard_submission-#        cm.first_name || ' ' || cm.last_name AS crew_name,
crewguard_submission-#        cm.crew_role,
crewguard_submission-#        COUNT(da.assignment_id) AS duties_assigned
crewguard_submission-# FROM crew_member cm
crewguard_submission-# LEFT JOIN duty_assignment da ON cm.crew_id = da.crew_id
crewguard_submission-# GROUP BY cm.crew_id, cm.employee_code, cm.first_name, cm.last_name, cm.crew_role
crewguard_submission-# ORDER BY duties_assigned DESC, cm.employee_code;
 employee_code |  crew_name   |   crew_role   | duties_assigned
---------------+--------------+---------------+-----------------
 EMP1001       | Arjun Menon  | CAPTAIN       |               4
 EMP1002       | Kavya Sharma | CAPTAIN       |               4
 EMP1003       | Rohan Kumar  | FIRST_OFFICER |               4
 EMP1005       | Vikram Joshi | CABIN_CREW    |               4
 EMP1004       | Sneha Patil  | FIRST_OFFICER |               2
 EMP1006       | Nisha Shah   | CABIN_CREW    |               2
 EMP1007       | Aditya Reddy | CAPTAIN       |               2
 EMP1008       | Meera Nair   | CABIN_CREW    |               0
(8 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

E4 - LEFT JOIN with IS NULL - crew holding no certification at all

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- E4  LEFT JOIN with IS NULL - crew holding no certification at all
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT cm.employee_code, cm.first_name, cm.last_name, cm.crew_role
crewguard_submission-# FROM crew_member cm
crewguard_submission-# LEFT JOIN certification c ON cm.crew_id = c.crew_id
crewguard_submission-# WHERE c.cert_id IS NULL;
 employee_code | first_name | last_name | crew_role
---------------+------------+-----------+------------
 EMP1005       | Vikram     | Joshi     | CABIN_CREW
 EMP1008       | Meera      | Nair      | CABIN_CREW
 EMP1006       | Nisha      | Shah      | CABIN_CREW
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

E5 - Self join - crew members based at the same airport

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- E5  Self join - crew members based at the same airport
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT a.employee_code AS crew_a,
crewguard_submission-#        b.employee_code AS crew_b,
crewguard_submission-#        a.home_base
crewguard_submission-# FROM crew_member a
crewguard_submission-# JOIN crew_member b
crewguard_submission-#   ON a.home_base = b.home_base
crewguard_submission-#  AND a.crew_id   < b.crew_id
crewguard_submission-# ORDER BY a.home_base, crew_a, crew_b;
 crew_a  | crew_b  | home_base
---------+---------+-----------
 EMP1001 | EMP1002 | BLR
 EMP1001 | EMP1003 | BLR
 EMP1001 | EMP1005 | BLR
 EMP1002 | EMP1003 | BLR
 EMP1002 | EMP1005 | BLR
 EMP1003 | EMP1005 | BLR
 EMP1004 | EMP1006 | DEL
(7 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

E6 - Join across the whole violation chain (the explanation query)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- E6  Join across the whole violation chain (the explanation query)
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT v.violation_id,
crewguard_submission-#        r.rule_name,
crewguard_submission-#        v.severity,
crewguard_submission-#        cm.employee_code,
crewguard_submission-#        cm.first_name || ' ' || cm.last_name AS crew_name,
crewguard_submission-#        f.flight_number,
crewguard_submission-#        da.duty_start_utc,
crewguard_submission-#        da.duty_end_utc,
crewguard_submission-#        va.contribution_value
crewguard_submission-# FROM violation v
crewguard_submission-# JOIN rule_definition      r  ON v.rule_code     = r.rule_code
crewguard_submission-# JOIN crew_member          cm ON v.crew_id       = cm.crew_id
crewguard_submission-# JOIN violation_assignment va ON v.violation_id  = va.violation_id
crewguard_submission-# JOIN duty_assignment      da ON va.assignment_id= da.assignment_id
crewguard_submission-# JOIN flight               f  ON da.flight_id    = f.flight_id
crewguard_submission-# ORDER BY v.violation_id, da.duty_start_utc;
 violation_id |            rule_name             | severity | employee_code |  crew_name  | flight_number |      duty_start_utc       |       duty_end_utc        | contribution_value
--------------+----------------------------------+----------+---------------+-------------+---------------+---------------------------+---------------------------+--------------------
            1 | Minimum Rest Between Duties      | HIGH     | EMP1001       | Arjun Menon | CG101         | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |               4.25
            1 | Minimum Rest Between Duties      | HIGH     | EMP1001       | Arjun Menon | CG102         | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |               4.25
            2 | Minimum Rest Between Duties      | HIGH     | EMP1003       | Rohan Kumar | CG101         | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |               4.25
            2 | Minimum Rest Between Duties      | HIGH     | EMP1003       | Rohan Kumar | CG102         | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |               4.25
            3 | Valid Type Certification Required | CRITICAL | EMP1004       | Sneha Patil | CG301         | 2026-09-03 06:00:00+05:30 | 2026-09-03 08:40:00+05:30 |               2.67
            3 | Valid Type Certification Required | CRITICAL | EMP1004       | Sneha Patil | CG302         | 2026-09-03 08:45:00+05:30 | 2026-09-03 11:10:00+05:30 |               2.42
(6 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART F : AGGREGATE FUNCTIONS, GROUP BY, HAVING -

F1 - Simple counts

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- F1  Simple counts
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT COUNT(*) AS total_crew         FROM crew_member;
 total_crew
------------
          8
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT COUNT(*) AS total_flights      FROM flight;
 total_flights
---------------
            10
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT COUNT(*) AS total_assignments   FROM duty_assignment;
 total_assignments
-------------------
                22
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT COUNT(*) AS active_rules       FROM rule_definition WHERE is_active;
 active_rules
--------------
            6
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

F2 - GROUP BY

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- F2  GROUP BY
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT crew_role, COUNT(*) AS crew_count
crewguard_submission-# FROM crew_member
crewguard_submission-# GROUP BY crew_role
crewguard_submission-# ORDER BY crew_count DESC;
   crew_role   | crew_count
---------------+------------
 CABIN_CREW    |          3
 CAPTAIN       |          3
 FIRST_OFFICER |          2
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT home_base, COUNT(*) AS crew_count
crewguard_submission-# FROM crew_member
crewguard_submission-# GROUP BY home_base
crewguard_submission-# ORDER BY home_base;
 home_base | crew_count
-----------+------------
 BLR       |          4
 BOM       |          1
 DEL       |          2
 MAA       |          1
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT type_code, COUNT(*) AS flights_operated
crewguard_submission-# FROM flight
crewguard_submission-# GROUP BY type_code
crewguard_submission-# ORDER BY flights_operated DESC;
 type_code | flights_operated
-----------+------------------
 A320      |                4
 A321      |                2
 B788      |                2
 B738      |                2
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

F3 - SUM / AVG / MIN / MAX on duty hours

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- F3  SUM / AVG / MIN / MAX on duty hours
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT cm.employee_code,
crewguard_submission-#        cm.first_name || ' ' || cm.last_name AS crew_name,
crewguard_submission-#        COUNT(*)                                                                    AS duties,
crewguard_submission-#        ROUND(SUM(EXTRACT(EPOCH FROM (da.duty_end_utc - da.duty_start_utc)) / 3600)::numeric, 2) AS total_duty_hours,
crewguard_submission-#        ROUND(AVG(EXTRACT(EPOCH FROM (da.duty_end_utc - da.duty_start_utc)) / 3600)::numeric, 2) AS avg_duty_hours,
crewguard_submission-#        ROUND(MAX(EXTRACT(EPOCH FROM (da.duty_end_utc - da.duty_start_utc)) / 3600)::numeric, 2) AS longest_duty_hours
crewguard_submission-# FROM duty_assignment da
crewguard_submission-# JOIN crew_member cm ON da.crew_id = cm.crew_id
crewguard_submission-# GROUP BY cm.crew_id, cm.employee_code, cm.first_name, cm.last_name
crewguard_submission-# ORDER BY total_duty_hours DESC;
 employee_code |  crew_name   | duties | total_duty_hours | avg_duty_hours | longest_duty_hours
---------------+--------------+--------+------------------+----------------+--------------------
 EMP1003       | Rohan Kumar  |      4 |            14.67 |           3.67 |               4.25
 EMP1001       | Arjun Menon  |      4 |            13.58 |           3.40 |               4.25
 EMP1005       | Vikram Joshi |      4 |            13.42 |           3.35 |               4.25
 EMP1002       | Kavya Sharma |      4 |            11.08 |           2.77 |               3.25
 EMP1007       | Aditya Reddy |      2 |            10.00 |           5.00 |               5.00
 EMP1006       | Nisha Shah   |      2 |             6.17 |           3.08 |               3.25
 EMP1004       | Sneha Patil  |      2 |             5.08 |           2.54 |               2.67
(7 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

F4 - HAVING - crew with more than three duties

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- F4  HAVING - crew with more than three duties
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT cm.employee_code, COUNT(*) AS duties
crewguard_submission-# FROM duty_assignment da
crewguard_submission-# JOIN crew_member cm ON da.crew_id = cm.crew_id
crewguard_submission-# GROUP BY cm.employee_code
crewguard_submission-# HAVING COUNT(*) > 3
crewguard_submission-# ORDER BY duties DESC;
 employee_code | duties
---------------+--------
 EMP1005       |      4
 EMP1002       |      4
 EMP1003       |      4
 EMP1001       |      4
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

F5 - HAVING on an aggregate expression - busiest sectors

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- F5  HAVING on an aggregate expression - busiest sectors
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT dep_airport, arr_airport, COUNT(*) AS flights
crewguard_submission-# FROM flight
crewguard_submission-# GROUP BY dep_airport, arr_airport
crewguard_submission-# HAVING COUNT(*) >= 1
crewguard_submission-# ORDER BY flights DESC, dep_airport;
 dep_airport | arr_airport | flights
-------------+-------------+---------
 BLR         | DEL         |       1
 BLR         | MAA         |       1
 BLR         | BOM         |       1
 BLR         | HYD         |       1
 BOM         | DXB         |       1
 BOM         | BLR         |       1
 DEL         | BLR         |       1
 DXB         | BOM         |       1
 HYD         | BLR         |       1
 MAA         | BLR         |       1
(10 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

F6 - Violations grouped by rule and by severity

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- F6  Violations grouped by rule and by severity
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT v.rule_code, r.rule_name, COUNT(*) AS violations
crewguard_submission-# FROM violation v
crewguard_submission-# JOIN rule_definition r ON v.rule_code = r.rule_code
crewguard_submission-# GROUP BY v.rule_code, r.rule_name
crewguard_submission-# ORDER BY violations DESC;
 rule_code |             rule_name             | violations
-----------+-----------------------------------+------------
 RST       | Minimum Rest Between Duties       |          2
 CER       | Valid Type Certification Required |          1
(2 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT severity, COUNT(*) AS violations
crewguard_submission-# FROM violation
crewguard_submission-# GROUP BY severity
crewguard_submission-# ORDER BY violations DESC;
 severity | violations
----------+------------
 HIGH     |          2
 CRITICAL |          1
(2 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

F7 - Aggregate with a conditional expression

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- F7  Aggregate with a conditional expression
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT COUNT(*)                                                AS total_certificates,
crewguard_submission-#        COUNT(*) FILTER (WHERE expires_on < CURRENT_DATE)        AS expired,
crewguard_submission-#        COUNT(*) FILTER (WHERE expires_on >= CURRENT_DATE)       AS valid
crewguard_submission-# FROM certification;
 total_certificates | expired | valid
--------------------+---------+-------
                  7 |       1 |     6
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART G : SUBQUERIES -

G1 - Sub-query with IN - crew who have at least one duty

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- G1  Sub-query with IN - crew who have at least one duty
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT employee_code, first_name, last_name
crewguard_submission-# FROM crew_member
crewguard_submission-# WHERE crew_id IN (SELECT crew_id FROM duty_assignment);
 employee_code | first_name | last_name
---------------+------------+-----------
 EMP1001       | Arjun      | Menon
 EMP1002       | Kavya      | Sharma
 EMP1003       | Rohan      | Kumar
 EMP1004       | Sneha      | Patil
 EMP1005       | Vikram     | Joshi
 EMP1006       | Nisha      | Shah
 EMP1007       | Aditya     | Reddy
(7 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

G2 - Sub-query with NOT IN - crew never rostered

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- G2  Sub-query with NOT IN - crew never rostered
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT employee_code, first_name, last_name
crewguard_submission-# FROM crew_member
crewguard_submission-# WHERE crew_id NOT IN (SELECT crew_id FROM duty_assignment);
 employee_code | first_name | last_name
---------------+------------+-----------
 EMP1008       | Meera      | Nair
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

G3 - EXISTS - crew with at least one recorded violation

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- G3  EXISTS - crew with at least one recorded violation
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT cm.employee_code, cm.first_name, cm.last_name
crewguard_submission-# FROM crew_member cm
crewguard_submission-# WHERE EXISTS (SELECT 1 FROM violation v WHERE v.crew_id = cm.crew_id);
 employee_code | first_name | last_name
---------------+------------+-----------
 EMP1001       | Arjun      | Menon
 EMP1003       | Rohan      | Kumar
 EMP1004       | Sneha      | Patil
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

G4 - NOT EXISTS - aircraft types never flown

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- G4  NOT EXISTS - aircraft types never flown
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT at.type_code, at.manufacturer
crewguard_submission-# FROM aircraft_type at
crewguard_submission-# WHERE NOT EXISTS (SELECT 1 FROM flight f WHERE f.type_code = at.type_code);
 type_code | manufacturer
-----------+--------------
(0 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

G5 - Scalar sub-query in the SELECT list

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- G5  Scalar sub-query in the SELECT list
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT f.flight_number,
crewguard_submission-#        f.dep_airport,
crewguard_submission-#        f.arr_airport,
crewguard_submission-#        (SELECT COUNT(*) FROM duty_assignment da WHERE da.flight_id = f.flight_id) AS crew_on_board
crewguard_submission-# FROM flight f
crewguard_submission-# ORDER BY crew_on_board DESC, f.flight_number;
 flight_number | dep_airport | arr_airport | crew_on_board
---------------+-------------+-------------+---------------
 CG101         | BLR         | DEL         |             3
 CG102         | DEL         | BLR         |             3
 CG201         | BLR         | BOM         |             3
 CG202         | BOM         | BLR         |             3
 CG301         | BLR         | MAA         |             2
 CG302         | MAA         | BLR         |             2
 CG501         | BLR         | HYD         |             2
 CG502         | HYD         | BLR         |             2
 CG401         | BOM         | DXB         |             1
 CG402         | DXB         | BOM         |             1
(10 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

G6 - Sub-query in WHERE with an aggregate (the average of a DATE column is computed over the number of days)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- G6  Sub-query in WHERE with an aggregate
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# --     (the average of a DATE column is computed over the number of days)
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT employee_code, first_name, last_name, hire_date
crewguard_submission-# FROM crew_member
crewguard_submission-# WHERE hire_date < (SELECT DATE '2000-01-01' + AVG(hire_date - DATE '2000-01-01')::INTEGER
crewguard_submission(#                    FROM crew_member)
crewguard_submission-# ORDER BY hire_date;
 employee_code | first_name | last_name | hire_date
---------------+------------+-----------+------------
 EMP1001       | Arjun      | Menon     | 2015-06-01
 EMP1007       | Aditya     | Reddy     | 2016-11-30
 EMP1002       | Kavya      | Sharma    | 2017-03-15
 EMP1003       | Rohan      | Kumar     | 2019-07-10
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

G7 - Correlated sub-query - the latest certificate of every crew member

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- G7  Correlated sub-query - the latest certificate of every crew member
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT c.cert_id, c.crew_id, c.type_code, c.issued_on, c.expires_on
crewguard_submission-# FROM certification c
crewguard_submission-# WHERE c.issued_on = (SELECT MAX(c2.issued_on)
crewguard_submission(#                      FROM certification c2
crewguard_submission(#                      WHERE c2.crew_id = c.crew_id)
crewguard_submission-# ORDER BY c.crew_id;
 cert_id | crew_id | type_code | issued_on  | expires_on
---------+---------+-----------+------------+------------
       2 |       1 | A321      | 2024-02-15 | 2027-02-14
       4 |       2 | B738      | 2025-01-05 | 2028-01-04
       5 |       3 | A320      | 2024-05-20 | 2027-05-19
       7 |       4 | B738      | 2023-08-12 | 2026-08-11
       8 |       7 | B788      | 2024-03-01 | 2027-02-28
(5 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

G8 - Derived table (sub-query in FROM)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- G8  Derived table (sub-query in FROM)
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT t.employee_code, t.total_duty_hours
crewguard_submission-# FROM (SELECT cm.employee_code,
crewguard_submission(#              ROUND(SUM(EXTRACT(EPOCH FROM (da.duty_end_utc - da.duty_start_utc)) / 3600)::numeric, 2) AS total_duty_hours
crewguard_submission(#       FROM duty_assignment da
crewguard_submission(#       JOIN crew_member cm ON da.crew_id = cm.crew_id
crewguard_submission(#       GROUP BY cm.employee_code) AS t
crewguard_submission-# WHERE t.total_duty_hours > 10
crewguard_submission-# ORDER BY t.total_duty_hours DESC;
 employee_code | total_duty_hours
---------------+------------------
 EMP1003       |           14.67
 EMP1001       |           13.58
 EMP1005       |           13.42
 EMP1002       |           11.08
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

G9 - Flight crew who operated an aircraft type they are not certified for (this is exactly what rule CER is meant to catch)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- G9  Flight crew who operated an aircraft type they are not certified for
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# --     (this is exactly what rule CER is meant to catch)
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT DISTINCT cm.employee_code,
crewguard_submission-#        cm.first_name || ' ' || cm.last_name AS crew_name,
crewguard_submission-#        f.type_code,
crewguard_submission-#        f.flight_number
crewguard_submission-# FROM duty_assignment da
crewguard_submission-# JOIN crew_member cm ON da.crew_id   = cm.crew_id
crewguard_submission-# JOIN flight      f  ON da.flight_id = f.flight_id
crewguard_submission-# WHERE NOT EXISTS (
crewguard_submission(#         SELECT 1
crewguard_submission(#         FROM certification c
crewguard_submission(#         WHERE c.crew_id   = da.crew_id
crewguard_submission(#           AND c.type_code = f.type_code
crewguard_submission(#           AND f.sched_dep_utc::date BETWEEN c.issued_on AND c.expires_on)
crewguard_submission-#   AND cm.crew_role <> 'CABIN_CREW'
crewguard_submission-# ORDER BY cm.employee_code, f.flight_number;
 employee_code |  crew_name  | type_code | flight_number
---------------+-------------+-----------+---------------
 EMP1003       | Rohan Kumar | B738      | CG201
 EMP1003       | Rohan Kumar | B738      | CG202
 EMP1004       | Sneha Patil | A321      | CG301
 EMP1004       | Sneha Patil | A321      | CG302
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART H : SET OPERATIONS -

H1 - UNION - every airport used as a departure or an arrival point

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- H1  UNION - every airport used as a departure or an arrival point
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT dep_airport AS airport_code, 'DEPARTURE' AS role FROM flight
crewguard_submission-# UNION
crewguard_submission-# SELECT arr_airport, 'ARRIVAL' FROM flight
crewguard_submission-# ORDER BY airport_code, role;
 airport_code |   role
--------------+-----------
 BLR          | ARRIVAL
 BLR          | DEPARTURE
 BOM          | ARRIVAL
 BOM          | DEPARTURE
 DEL          | ARRIVAL
 DEL          | DEPARTURE
 DXB          | ARRIVAL
 DXB          | DEPARTURE
 HYD          | ARRIVAL
 HYD          | DEPARTURE
 MAA          | ARRIVAL
 MAA          | DEPARTURE
(12 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

H2 - INTERSECT - airports that are both a crew base and a flight origin

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- H2  INTERSECT - airports that are both a crew base and a flight origin
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT home_base   FROM crew_member
crewguard_submission-# INTERSECT
crewguard_submission-# SELECT dep_airport FROM flight;
 home_base
-----------
 MAA
 BLR
 DEL
 BOM
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

H3 - EXCEPT - airports in the master table never used by any flight

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- H3  EXCEPT - airports in the master table never used by any flight
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT iata_code FROM airport
crewguard_submission-# EXCEPT
crewguard_submission-# SELECT dep_airport FROM flight
crewguard_submission-# EXCEPT
crewguard_submission-# SELECT arr_airport FROM flight;
 iata_code
-----------
(0 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART I : VIEWS -

I1 - Operational roster view

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- I1  Operational roster view
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE OR REPLACE VIEW v_duty_roster AS
crewguard_submission-# SELECT da.assignment_id,
crewguard_submission-#        cm.crew_id,
crewguard_submission-#        cm.employee_code,
crewguard_submission-#        cm.first_name || ' ' || cm.last_name AS crew_name,
crewguard_submission-#        da.assigned_role,
crewguard_submission-#        f.flight_number,
crewguard_submission-#        f.dep_airport,
crewguard_submission-#        f.arr_airport,
crewguard_submission-#        f.sched_dep_utc,
crewguard_submission-#        f.sched_arr_utc,
crewguard_submission-#        da.duty_start_utc,
crewguard_submission-#        da.duty_end_utc,
crewguard_submission-#        ROUND((EXTRACT(EPOCH FROM (da.duty_end_utc - da.duty_start_utc)) / 3600)::numeric, 2) AS duty_hours
crewguard_submission-# FROM duty_assignment da
crewguard_submission-# JOIN crew_member cm ON da.crew_id   = cm.crew_id
crewguard_submission-# JOIN flight      f  ON da.flight_id = f.flight_id;
CREATE VIEW
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT * FROM v_duty_roster ORDER BY duty_start_utc LIMIT 10;
 assignment_id | crew_id | employee_code |  crew_name   | assigned_role | flight_number | dep_airport | arr_airport |       sched_dep_utc       |       sched_arr_utc       |       duty_start_utc      |       duty_end_utc        | duty_hours
---------------+---------+---------------+--------------+---------------+---------------+-------------+-------------+---------------------------+---------------------------+---------------------------+---------------------------+------------
             1 |       1 | EMP1001       | Arjun Menon  | CAPTAIN       | CG101         | BLR         | DEL         | 2026-09-01 06:30:00+05:30 | 2026-09-01 09:15:00+05:30 | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |       4.25
             9 |       3 | EMP1003       | Rohan Kumar  | FIRST_OFFICER | CG101         | BLR         | DEL         | 2026-09-01 06:30:00+05:30 | 2026-09-01 09:15:00+05:30 | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |       4.25
            15 |       5 | EMP1005       | Vikram Joshi | CABIN_CREW    | CG101         | BLR         | DEL         | 2026-09-01 06:30:00+05:30 | 2026-09-01 09:15:00+05:30 | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |       4.25
             2 |       1 | EMP1001       | Arjun Menon  | CAPTAIN       | CG102         | DEL         | BLR         | 2026-09-01 11:00:00+05:30 | 2026-09-01 13:45:00+05:30 | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |       4.25
            10 |       3 | EMP1003       | Rohan Kumar  | FIRST_OFFICER | CG102         | DEL         | BLR         | 2026-09-01 11:00:00+05:30 | 2026-09-01 13:45:00+05:30 | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |       4.25
            16 |       5 | EMP1005       | Vikram Joshi | CABIN_CREW    | CG102         | DEL         | BLR         | 2026-09-01 11:00:00+05:30 | 2026-09-01 13:45:00+05:30 | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |       4.25
            19 |       6 | EMP1006       | Nisha Shah   | CABIN_CREW    | CG201         | BLR         | BOM         | 2026-09-02 07:30:00+05:30 | 2026-09-02 09:15:00+05:30 | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30 |       3.25
             5 |       2 | EMP1002       | Kavya Sharma | CAPTAIN       | CG201         | BLR         | BOM         | 2026-09-02 07:30:00+05:30 | 2026-09-02 09:15:00+05:30 | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30 |       3.25
            11 |       3 | EMP1003       | Rohan Kumar  | FIRST_OFFICER | CG201         | BLR         | BOM         | 2026-09-02 07:30:00+05:30 | 2026-09-02 09:15:00+05:30 | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30 |       3.25
             6 |       2 | EMP1002       | Kavya Sharma | CAPTAIN       | CG202         | BOM         | BLR         | 2026-09-02 10:30:00+05:30 | 2026-09-02 12:15:00+05:30 | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30 |       2.92
(10 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

I2 - Compliance report view

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- I2  Compliance report view
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE OR REPLACE VIEW v_violation_report AS
crewguard_submission-# SELECT v.violation_id,
crewguard_submission-#        vr.run_id,
crewguard_submission-#        vr.period_start,
crewguard_submission-#        vr.period_end,
crewguard_submission-#        u.username          AS triggered_by,
crewguard_submission-#        r.rule_code,
crewguard_submission-#        r.rule_name,
crewguard_submission-#        v.severity,
crewguard_submission-#        cm.employee_code,
crewguard_submission-#        cm.first_name || ' ' || cm.last_name AS crew_name,
crewguard_submission-#        v.window_start,
crewguard_submission-#        v.window_end,
crewguard_submission-#        v.observed_value,
crewguard_submission-#        v.threshold_value,
crewguard_submission-#        v.evidence_doc_id
crewguard_submission-# FROM violation v
crewguard_submission-# JOIN validation_run  vr ON v.run_id    = vr.run_id
crewguard_submission-# JOIN rule_definition r  ON v.rule_code = r.rule_code
crewguard_submission-# JOIN crew_member     cm ON v.crew_id   = cm.crew_id
crewguard_submission-# LEFT JOIN app_user   u  ON vr.triggered_by = u.user_id;
CREATE VIEW
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT * FROM v_violation_report ORDER BY violation_id;
 violation_id | run_id | period_start | period_end | triggered_by | rule_code |          rule_name          | severity | employee_code |  crew_name   |       window_start        |        window_end         | observed_value | threshold_value | evidence_doc_id
--------------+--------+--------------+------------+--------------+-----------+-----------------------------+----------+---------------+--------------+---------------------------+---------------------------+----------------+-----------------+-----------------
            1 |      1 | 2026-09-01   | 2026-09-07 | audit_meera  | REST      | Minimum Rest Between Duties | HIGH     | EMP1001       | Arjun Menon  | 2026-09-01 09:45:00+05:30 | 2026-09-01 10:08:00+05:30 |           0.25 |           10.00 | DOC-2026-0001
            2 |      1 | 2026-09-01   | 2026-09-07 | audit_meera  | REST      | Minimum Rest Between Duties | HIGH     | EMP1003       | Rohan Kumar  | 2026-09-01 09:45:00+05:30 | 2026-09-01 10:08:00+05:30 |           0.25 |           10.00 | DOC-2026-0002
            3 |      1 | 2026-09-01   | 2026-09-07 | audit_meera  | CER       | Valid Type Certification Required | CRITICAL | EMP1004       | Sneha Patil  | 2026-09-03 06:00:00+05:30 | 2026-09-03 11:10:00+05:30 |                |                 | DOC-2026-0003
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

I3 - Querying a view like a table

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- I3  Querying a view like a table
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT employee_code, COUNT(*) AS violations
crewguard_submission-# FROM v_violation_report
crewguard_submission-# GROUP BY employee_code
crewguard_submission-# ORDER BY violations DESC;
 employee_code | violations
---------------+------------
 EMP1004       |          1
 EMP1003       |          1
 EMP1001       |          1
(3 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART J : FUNCTIONS -

J1 - Scalar function - duty length in minutes

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- J1  Scalar function - duty length in minutes
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE OR REPLACE FUNCTION fn_duty_minutes(p_start TIMESTAMPTZ, p_end TIMESTAMPTZ)
crewguard_submission-# RETURNS INTEGER
crewguard_submission-# LANGUAGE plpgsql
crewguard_submission-# IMMUTABLE
crewguard_submission-# AS $$
crewguard_submission$# BEGIN
crewguard_submission$#     RETURN (EXTRACT(EPOCH FROM (p_end - p_start)) / 60)::INTEGER;
crewguard_submission$# END;
crewguard_submission$# $$;
CREATE FUNCTION
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT assignment_id,
crewguard_submission-#        duty_start_utc,
crewguard_submission-#        duty_end_utc,
crewguard_submission-#        fn_duty_minutes(duty_start_utc, duty_end_utc) AS duty_minutes
crewguard_submission-# FROM duty_assignment
crewguard_submission-# ORDER BY assignment_id
crewguard_submission-# LIMIT 10;
 assignment_id |      duty_start_utc       |       duty_end_utc        | duty_minutes
---------------+---------------------------+---------------------------+--------------
             1 | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |          255
             2 | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |          255
             3 | 2026-09-03 06:00:00+05:30 | 2026-09-03 08:40:00+05:30 |          160
             4 | 2026-09-03 08:45:00+05:30 | 2026-09-03 11:10:00+05:30 |          145
             5 | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30 |          195
             6 | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30 |          175
             7 | 2026-09-06 06:30:00+05:30 | 2026-09-06 09:00:00+05:30 |          150
             8 | 2026-09-06 09:05:00+05:30 | 2026-09-06 11:30:00+05:30 |          145
             9 | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |          255
            10 | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |          255
(10 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

J2 - Aggregate function - total duty hours of a crew member in a window

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- J2  Aggregate function - total duty hours of a crew member in a window
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE OR REPLACE FUNCTION fn_crew_duty_hours(p_crew_id INTEGER,
crewguard_submission(#                                                 p_from    TIMESTAMPTZ,
crewguard_submission(#                                                 p_to      TIMESTAMPTZ)
crewguard_submission-# RETURNS NUMERIC
crewguard_submission-# LANGUAGE plpgsql
crewguard_submission-# AS $$
crewguard_submission$# DECLARE
crewguard_submission$#     v_hours NUMERIC(8,2);
crewguard_submission$# BEGIN
crewguard_submission$#     SELECT COALESCE(ROUND(SUM(EXTRACT(EPOCH FROM (duty_end_utc - duty_start_utc)) / 3600)::numeric, 2), 0)
crewguard_submission$#       INTO v_hours
crewguard_submission$#       FROM duty_assignment
crewguard_submission$#      WHERE crew_id = p_crew_id
crewguard_submission$#        AND duty_start_utc >= p_from
crewguard_submission$#        AND duty_start_utc <  p_to;
crewguard_submission$#     RETURN v_hours;
crewguard_submission$# END;
crewguard_submission$# $$;
CREATE FUNCTION
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT cm.employee_code,
crewguard_submission-#        fn_crew_duty_hours(cm.crew_id, '2026-09-01', '2026-09-08') AS duty_hours_week1
crewguard_submission-# FROM crew_member cm
crewguard_submission-# ORDER BY duty_hours_week1 DESC;
 employee_code | duty_hours_week1
---------------+------------------
 EMP1003       |            14.67
 EMP1001       |            13.58
 EMP1005       |            13.42
 EMP1002       |            11.08
 EMP1007       |            10.00
 EMP1006       |             6.17
 EMP1004       |             5.08
 EMP1008       |             0.00
(8 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

J3 - Table valued function - the roster of one crew member

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- J3  Table valued function - the roster of one crew member
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE OR REPLACE FUNCTION fn_crew_roster(p_crew_id INTEGER)
crewguard_submission-# RETURNS TABLE (flight_number  VARCHAR(10),
crewguard_submission(#                sector        TEXT,
crewguard_submission(#                duty_start_utc TIMESTAMPTZ,
crewguard_submission(#                duty_end_utc   TIMESTAMPTZ,
crewguard_submission(#                duty_hours     NUMERIC)
crewguard_submission-# LANGUAGE plpgsql
crewguard_submission-# AS $$
crewguard_submission$# BEGIN
crewguard_submission$#     RETURN QUERY
crewguard_submission$#     SELECT f.flight_number,
crewguard_submission$#            f.dep_airport || ' -> ' || f.arr_airport,
crewguard_submission$#            da.duty_start_utc,
crewguard_submission$#            da.duty_end_utc,
crewguard_submission$#            ROUND((EXTRACT(EPOCH FROM (da.duty_end_utc - da.duty_start_utc)) / 3600)::numeric, 2)
crewguard_submission$#     FROM duty_assignment da
crewguard_submission$#     JOIN flight f ON da.flight_id = f.flight_id
crewguard_submission$#     WHERE da.crew_id = p_crew_id
crewguard_submission$#     ORDER BY da.duty_start_utc;
crewguard_submission$# END;
crewguard_submission$# $$;
CREATE FUNCTION
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT * FROM fn_crew_roster(1);
 flight_number |   sector    |      duty_start_utc       |       duty_end_utc        | duty_hours
---------------+-------------+---------------------------+---------------------------+------------
 CG101         | BLR -> DEL | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |       4.25
 CG102         | DEL -> BLR | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |       4.25
 CG301         | BLR -> MAA | 2026-09-03 06:00:00+05:30 | 2026-09-03 08:40:00+05:30 |       2.67
 CG302         | MAA -> BLR | 2026-09-03 08:45:00+05:30 | 2026-09-03 11:10:00+05:30 |       2.42
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART K : TRIGGERS -

K1 - Keep validation_run.violations_found synchronised automatically

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- K1  Keep validation_run.violations_found synchronised automatically
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE OR REPLACE FUNCTION trg_fn_sync_violation_count()
crewguard_submission-# RETURNS TRIGGER
crewguard_submission-# LANGUAGE plpgsql
crewguard_submission-# AS $$
crewguard_submission$# BEGIN
crewguard_submission$#     IF TG_OP = 'INSERT' THEN
crewguard_submission$#         UPDATE validation_run
crewguard_submission$#         SET violations_found = violations_found + 1
crewguard_submission$#         WHERE run_id = NEW.run_id;
crewguard_submission$#         RETURN NEW;
crewguard_submission$#     ELSIF TG_OP = 'DELETE' THEN
crewguard_submission$#         UPDATE validation_run
crewguard_submission$#         SET violations_found = GREATEST(violations_found - 1, 0)
crewguard_submission$#         WHERE run_id = OLD.run_id;
crewguard_submission$#         RETURN OLD;
crewguard_submission$#     END IF;
crewguard_submission$#     RETURN NULL;
crewguard_submission$# END;
crewguard_submission$# $$;
CREATE FUNCTION
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# DROP TRIGGER IF EXISTS trg_violation_count ON violation;
NOTICE:  trigger "trg_violation_count" for relation "violation" does not exist, skipping
DROP TRIGGER
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TRIGGER trg_violation_count
crewguard_submission-# AFTER INSERT OR DELETE ON violation
crewguard_submission-# FOR EACH ROW
crewguard_submission-# EXECUTE FUNCTION trg_fn_sync_violation_count();
CREATE TRIGGER
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

K2 - Stamp finished_at automatically when a run is closed

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- K2  Stamp finished_at automatically when a run is closed
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE OR REPLACE FUNCTION trg_fn_stamp_run_finish()
crewguard_submission-# RETURNS TRIGGER
crewguard_submission-# LANGUAGE plpgsql
crewguard_submission-# AS $$
crewguard_submission$# BEGIN
crewguard_submission$#     IF NEW.status IN ('COMPLETED', 'FAILED') AND NEW.finished_at IS NULL THEN
crewguard_submission$#         NEW.finished_at := NOW();
crewguard_submission$#     END IF;
crewguard_submission$#     RETURN NEW;
crewguard_submission$# END;
crewguard_submission$# $$;
CREATE FUNCTION
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# DROP TRIGGER IF EXISTS trg_run_finish ON validation_run;
NOTICE:  trigger "trg_run_finish" for relation "validation_run" does not exist, skipping
DROP TRIGGER
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE TRIGGER trg_run_finish
crewguard_submission-# BEFORE UPDATE ON validation_run
crewguard_submission-# FOR EACH ROW
crewguard_submission-# EXECUTE FUNCTION trg_fn_stamp_run_finish();
CREATE TRIGGER
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

K3 - Demonstration of trigger K1 (counter goes up on INSERT)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- K3  Demonstration of trigger K1 (counter goes up on INSERT)
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT run_id, status, violations_found FROM validation_run WHERE run_id = 2;
 run_id |  status   | violations_found
--------+-----------+------------------
      2 | COMPLETED |                0
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO violation (run_id, rule_code, crew_id, severity,
crewguard_submission(#                         window_start, window_end, observed_value, threshold_value, evidence_doc_id)
crewguard_submission-# VALUES (2, 'FDP', 7, 'CRITICAL',
crewguard_submission(#         '2026-09-04 17:00+00', '2026-09-05 05:00+00', 12.00, 13.00, 'DOC-2026-0004')
crewguard_submission-# RETURNING violation_id, run_id, rule_code;
 violation_id | run_id | rule_code
--------------+--------+-----------
            4 |      2 | FDP
(1 row)

INSERT 0 1
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT run_id, status, violations_found FROM validation_run WHERE run_id = 2;
 run_id |  status   | violations_found
--------+-----------+------------------
      2 | COMPLETED |                1
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

K4 - Demonstration of trigger K1 (counter goes down on DELETE, and ON DELETE CASCADE removes the dependent explanation rows)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- K4  Demonstration of trigger K1 (counter goes down on DELETE,
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# --     and ON DELETE CASCADE removes the dependent explanation rows)
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT COUNT(*) AS va_rows_before FROM violation_assignment WHERE violation_id = 3;
 va_rows_before
----------------
              2
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# DELETE FROM violation WHERE violation_id = 3
crewguard_submission-# RETURNING violation_id, run_id, rule_code;
 violation_id | run_id | rule_code
--------------+--------+-----------
            3 |      1 | CER
(1 row)

DELETE 1
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT COUNT(*) AS va_rows_after FROM violation_assignment WHERE violation_id = 3;
 va_rows_after
---------------
             0
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT run_id, violations_found FROM validation_run WHERE run_id = 1;
 run_id | violations_found
--------+------------------
      1 |                2
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

K5 - Demonstration of trigger K2 (finished_at filled in automatically)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- K5  Demonstration of trigger K2 (finished_at filled in automatically)
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT run_id, status, finished_at FROM validation_run WHERE run_id = 3;
 run_id | status | finished_at
--------+--------+-------------
      3 | STALE  |
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# UPDATE validation_run SET status = 'COMPLETED' WHERE run_id = 3
crewguard_submission-# RETURNING run_id, status, finished_at;
 run_id |  status   |           finished_at
--------+-----------+----------------------------------
      3 | COMPLETED | 2026-10-04 17:35:47.159776+05:30
(1 row)

UPDATE 1
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART L : STORED PROCEDURE (THE COMPLIANCE VALIDATOR) -

L0 - Setup: create procedure

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE OR REPLACE PROCEDURE sp_validate_rest_periods(p_user_id      INTEGER,
crewguard_submission(#                                                          p_period_start DATE,
crewguard_submission(#                                                          p_period_end   DATE)
crewguard_submission-# LANGUAGE plpgsql
crewguard_submission-# AS $$
crewguard_submission$# DECLARE
crewguard_submission$#     v_run_id       INTEGER;
crewguard_submission$#     v_threshold    NUMERIC(8,2);
crewguard_submission$#     v_severity     VARCHAR(10);
crewguard_submission$#     v_violation_id INTEGER;
crewguard_submission$#     v_count        INTEGER := 0;
crewguard_submission$#     r              RECORD;
crewguard_submission$# BEGIN
crewguard_submission$#     -- 1. read the rule that is being enforced
crewguard_submission$#     SELECT threshold_value, severity
crewguard_submission$#       INTO v_threshold, v_severity
crewguard_submission$#       FROM rule_definition
crewguard_submission$#      WHERE rule_code = 'RST' AND is_active;
crewguard_submission$#
crewguard_submission$#     IF NOT FOUND THEN
crewguard_submission$#         RAISE EXCEPTION 'Active rule RST is not defined';
crewguard_submission$#     END IF;
crewguard_submission$#
crewguard_submission$#     -- 2. open a validation run
crewguard_submission$#     INSERT INTO validation_run (triggered_by, period_start, period_end, status)
crewguard_submission$#     VALUES (p_user_id, p_period_start, p_period_end, 'RUNNING')
crewguard_submission$#     RETURNING run_id INTO v_run_id;
crewguard_submission$#
crewguard_submission$#     -- 3. scan consecutive duties of every crew member
crewguard_submission$#     FOR r IN
crewguard_submission$#         WITH ordered AS (
crewguard_submission$#             SELECT da.assignment_id,
crewguard_submission$#                    da.crew_id,
crewguard_submission$#                    da.duty_start_utc,
crewguard_submission$#                    da.duty_end_utc,
crewguard_submission$#                    LAG(da.assignment_id)  OVER w AS prev_assignment_id,
crewguard_submission$#                    LAG(da.duty_start_utc) OVER w AS prev_start,
crewguard_submission$#                    LAG(da.duty_end_utc)   OVER w AS prev_end
crewguard_submission$#             FROM duty_assignment da
crewguard_submission$#             WHERE da.duty_start_utc >= p_period_start::timestamptz
crewguard_submission$#               AND da.duty_start_utc <  (p_period_end + 1)::timestamptz
crewguard_submission$#             WINDOW w AS (PARTITION BY da.crew_id ORDER BY da.duty_start_utc)
crewguard_submission$#         )
crewguard_submission$#         SELECT o.*,
```

L0 - Setup: create procedure (contd.)

```
crewguard_submission$#                ROUND((EXTRACT(EPOCH FROM (o.duty_start_utc - o.prev_end)) / 3600)::numeric, 2) AS rest_hours
crewguard_submission$#         FROM ordered o
crewguard_submission$#         WHERE o.prev_end IS NOT NULL
crewguard_submission$#           AND (EXTRACT(EPOCH FROM (o.duty_start_utc - o.prev_end)) / 3600) < v_threshold
crewguard_submission$#         ORDER BY o.crew_id, o.duty_start_utc
crewguard_submission$#     LOOP
crewguard_submission$#         -- 3a. record the violation
crewguard_submission$#         INSERT INTO violation (run_id, rule_code, crew_id, severity,
crewguard_submission$#                                window_start, window_end,
crewguard_submission$#                                observed_value, threshold_value, evidence_doc_id)
crewguard_submission$#         VALUES (v_run_id, 'RST', r.crew_id, v_severity,
crewguard_submission$#                 r.prev_end, r.duty_start_utc,
crewguard_submission$#                 r.rest_hours, v_threshold,
crewguard_submission$#                 'RUN' || LPAD(v_run_id::text, 4, '0') || '-V' || LPAD((v_count + 1)::text, 4, '0'))
crewguard_submission$#         RETURNING violation_id INTO v_violation_id;
crewguard_submission$#
crewguard_submission$#         -- 3b. explain it: the two duties that produced the short rest
crewguard_submission$#         INSERT INTO violation_assignment (violation_id, assignment_id, contribution_value)
crewguard_submission$#         VALUES (v_violation_id, r.prev_assignment_id,
crewguard_submission$#                 ROUND((EXTRACT(EPOCH FROM (r.prev_end - r.prev_start)) / 3600)::numeric, 2)),
crewguard_submission$#                (v_violation_id, r.assignment_id,
crewguard_submission$#                 ROUND((EXTRACT(EPOCH FROM (r.duty_end_utc - r.duty_start_utc)) / 3600)::numeric, 2));
crewguard_submission$#
crewguard_submission$#         v_count := v_count + 1;
crewguard_submission$#     END LOOP;
crewguard_submission$#
crewguard_submission$#     -- 4. close the run
crewguard_submission$#     UPDATE validation_run
crewguard_submission$#     SET status           = 'COMPLETED',
crewguard_submission$#         finished_at      = NOW(),
crewguard_submission$#         violations_found = (SELECT COUNT(*) FROM violation WHERE run_id = v_run_id)
crewguard_submission$#     WHERE run_id = v_run_id;
crewguard_submission$#
crewguard_submission$#     RAISE NOTICE 'Validation run % completed: % violation(s) of rule RST detected',
crewguard_submission$#                  v_run_id, v_count;
crewguard_submission$# END;
crewguard_submission$# $$;
CREATE PROCEDURE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

L1 - Execute the validator for the first week of September 2026

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- L1  Execute the validator for the first week of September 2026
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CALL sp_validate_rest_periods(3, DATE '2026-09-01', DATE '2026-09-07');
NOTICE:  Validation run 4 completed: 11 violation(s) of rule RST detected
CALL
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

L2 - The run that was just created

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- L2  The run that was just created
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT run_id, triggered_by, period_start, period_end, status, violations_found
crewguard_submission-# FROM validation_run
crewguard_submission-# ORDER BY run_id DESC
crewguard_submission-# LIMIT 1;
 run_id | triggered_by | period_start | period_end |  status   | violations_found
--------+--------------+--------------+------------+-----------+------------------
      4 |            3 | 2026-09-01   | 2026-09-07 | COMPLETED |               11
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

L3 - The violations the procedure detected

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- L3  The violations the procedure detected
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT v.violation_id, cm.employee_code, v.severity,
crewguard_submission-#        v.window_start, v.window_end,
crewguard_submission-#        v.observed_value AS rest_hours,
crewguard_submission-#        v.threshold_value AS required_hours
crewguard_submission-# FROM violation v
crewguard_submission-# JOIN crew_member cm ON v.crew_id = cm.crew_id
crewguard_submission-# WHERE v.run_id = (SELECT MAX(run_id) FROM validation_run)
crewguard_submission-# ORDER BY cm.employee_code, v.window_start;
 violation_id | employee_code | severity |       window_start        |        window_end         | rest_hours | required_hours
--------------+---------------+----------+---------------------------+---------------------------+------------+----------------
            5 | EMP1001       | HIGH     | 2026-09-01 09:45:00+05:30 | 2026-09-01 10:00:00+05:30 |       0.25 |          10.00
            6 | EMP1001       | HIGH     | 2026-09-03 08:40:00+05:30 | 2026-09-03 08:45:00+05:30 |       0.08 |          10.00
            7 | EMP1002       | HIGH     | 2026-09-02 09:45:00+05:30 | 2026-09-02 09:50:00+05:30 |       0.08 |          10.00
            8 | EMP1002       | HIGH     | 2026-09-06 09:00:00+05:30 | 2026-09-06 09:05:00+05:30 |       0.08 |          10.00
            9 | EMP1003       | HIGH     | 2026-09-01 09:45:00+05:30 | 2026-09-01 10:00:00+05:30 |       0.25 |          10.00
           10 | EMP1003       | HIGH     | 2026-09-02 09:45:00+05:30 | 2026-09-02 09:50:00+05:30 |       0.08 |          10.00
           11 | EMP1004       | HIGH     | 2026-09-03 08:40:00+05:30 | 2026-09-03 08:45:00+05:30 |       0.08 |          10.00
           12 | EMP1005       | HIGH     | 2026-09-01 09:45:00+05:30 | 2026-09-01 10:00:00+05:30 |       0.25 |          10.00
           13 | EMP1005       | HIGH     | 2026-09-06 09:00:00+05:30 | 2026-09-06 09:05:00+05:30 |       0.08 |          10.00
           14 | EMP1006       | HIGH     | 2026-09-02 09:45:00+05:30 | 2026-09-02 09:50:00+05:30 |       0.08 |          10.00
           15 | EMP1007       | HIGH     | 2026-09-05 03:30:00+05:30 | 2026-09-05 05:30:00+05:30 |       2.00 |          10.00
(11 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

L4 - The explanation: which duties caused each violation

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- L4  The explanation: which duties caused each violation
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT v.violation_id,
crewguard_submission-#        cm.employee_code,
crewguard_submission-#        f.flight_number,
crewguard_submission-#        da.duty_start_utc,
crewguard_submission-#        da.duty_end_utc,
crewguard_submission-#        va.contribution_value AS duty_hours
crewguard_submission-# FROM violation v
crewguard_submission-# JOIN crew_member          cm ON v.crew_id        = cm.crew_id
crewguard_submission-# JOIN violation_assignment va ON v.violation_id   = va.violation_id
crewguard_submission-# JOIN duty_assignment      da ON va.assignment_id = da.assignment_id
crewguard_submission-# JOIN flight               f  ON da.flight_id     = f.flight_id
crewguard_submission-# WHERE v.run_id = (SELECT MAX(run_id) FROM validation_run)
crewguard_submission-# ORDER BY v.violation_id, da.duty_start_utc;
 violation_id | employee_code | flight_number |      duty_start_utc       |       duty_end_utc        | duty_hours
--------------+---------------+---------------+---------------------------+---------------------------+------------
            5 | EMP1001       | CG101         | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |       4.25
            5 | EMP1001       | CG102         | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |       4.25
            6 | EMP1001       | CG301         | 2026-09-03 06:00:00+05:30 | 2026-09-03 08:40:00+05:30 |       2.67
            6 | EMP1001       | CG302         | 2026-09-03 08:45:00+05:30 | 2026-09-03 11:10:00+05:30 |       2.42
            7 | EMP1002       | CG201         | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30 |       3.25
            7 | EMP1002       | CG202         | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30 |       2.92
            8 | EMP1002       | CG501         | 2026-09-06 06:30:00+05:30 | 2026-09-06 09:00:00+05:30 |       2.50
            8 | EMP1002       | CG502         | 2026-09-06 09:05:00+05:30 | 2026-09-06 11:30:00+05:30 |       2.42
            9 | EMP1003       | CG101         | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |       4.25
            9 | EMP1003       | CG102         | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |       4.25
           10 | EMP1003       | CG201         | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30 |       3.25
           10 | EMP1003       | CG202         | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30 |       2.92
           11 | EMP1004       | CG301         | 2026-09-03 06:00:00+05:30 | 2026-09-03 08:40:00+05:30 |       2.67
           11 | EMP1004       | CG302         | 2026-09-03 08:45:00+05:30 | 2026-09-03 11:10:00+05:30 |       2.42
           12 | EMP1005       | CG101         | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |       4.25
           12 | EMP1005       | CG102         | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 |       4.25
           13 | EMP1005       | CG501         | 2026-09-06 06:30:00+05:30 | 2026-09-06 09:00:00+05:30 |       2.50
           13 | EMP1005       | CG502         | 2026-09-06 09:05:00+05:30 | 2026-09-06 11:30:00+05:30 |       2.42
           14 | EMP1006       | CG201         | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30 |       3.25
           14 | EMP1006       | CG202         | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30 |       2.92
           15 | EMP1007       | CG401         | 2026-09-04 22:30:00+05:30 | 2026-09-05 03:30:00+05:30 |       5.00
           15 | EMP1007       | CG402         | 2026-09-05 05:30:00+05:30 | 2026-09-05 10:30:00+05:30 |       5.00
(22 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART M : COMMON TABLE EXPRESSIONS AND WINDOW FUNCTIONS -

M1 - Simple CTE - crew whose duty hours exceed the fleet average

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- M1  Simple CTE - crew whose duty hours exceed the fleet average
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# WITH crew_hours AS (
crewguard_submission(#     SELECT cm.crew_id,
crewguard_submission(#            cm.employee_code,
crewguard_submission(#            ROUND(SUM(EXTRACT(EPOCH FROM (da.duty_end_utc - da.duty_start_utc)) / 3600)::numeric, 2) AS total_hours
crewguard_submission(#     FROM duty_assignment da
crewguard_submission(#     JOIN crew_member cm ON da.crew_id = cm.crew_id
crewguard_submission(#     GROUP BY cm.crew_id, cm.employee_code
crewguard_submission(# )
crewguard_submission-# SELECT employee_code, total_hours
crewguard_submission-# FROM crew_hours
crewguard_submission-# WHERE total_hours > (SELECT AVG(total_hours) FROM crew_hours)
crewguard_submission-# ORDER BY total_hours DESC;
 employee_code | total_hours
---------------+-------------
 EMP1003       |       14.67
 EMP1001       |       13.58
 EMP1005       |       13.42
 EMP1002       |       11.08
(4 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

M2 - Multiple CTEs chained together - violation load per crew member

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- M2  Multiple CTEs chained together — violation load per crew member
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# WITH duty_load AS (
crewguard_submission(#     SELECT crew_id, COUNT(*) AS duties
crewguard_submission(#     FROM duty_assignment
crewguard_submission(#     GROUP BY crew_id
crewguard_submission(# ),
crewguard_submission-# violation_load AS (
crewguard_submission(#     SELECT crew_id, COUNT(*) AS violations
crewguard_submission(#     FROM violation
crewguard_submission(#     GROUP BY crew_id
crewguard_submission(# )
crewguard_submission-# SELECT cm.employee_code,
crewguard_submission-#        COALESCE(d.duties, 0)     AS duties,
crewguard_submission-#        COALESCE(v.violations, 0) AS violations,
crewguard_submission-#        ROUND(COALESCE(v.violations, 0)::numeric / NULLIF(d.duties, 0), 2) AS violations_per_duty
crewguard_submission-# FROM crew_member cm
crewguard_submission-# LEFT JOIN duty_load      d ON cm.crew_id = d.crew_id
crewguard_submission-# LEFT JOIN violation_load v ON cm.crew_id = v.crew_id
crewguard_submission-# ORDER BY violations DESC, duties DESC;
 employee_code | duties | violations | violations_per_duty
---------------+--------+------------+---------------------
 EMP1003       |      4 |          3 |                0.75
 EMP1001       |      4 |          3 |                0.75
 EMP1005       |      4 |          2 |                0.50
 EMP1002       |      4 |          2 |                0.50
 EMP1007       |      2 |          2 |                1.00
 EMP1004       |      2 |          1 |                0.50
 EMP1006       |      2 |          1 |                0.50
 EMP1008       |      0 |          0 |
(8 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

M3 - Window functions - rank crew by duty hours

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- M3  Window functions — rank crew by duty hours
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT cm.employee_code,
crewguard_submission-#        ROUND(SUM(EXTRACT(EPOCH FROM (da.duty_end_utc - da.duty_start_utc)) / 3600)::numeric, 2) AS total_hours,
crewguard_submission-#        RANK() OVER (ORDER BY SUM(da.duty_end_utc - da.duty_start_utc) DESC) AS duty_rank
crewguard_submission-# FROM duty_assignment da
crewguard_submission-# JOIN crew_member cm ON da.crew_id = cm.crew_id
crewguard_submission-# GROUP BY cm.employee_code
crewguard_submission-# ORDER BY duty_rank;
 employee_code | total_hours | duty_rank
---------------+-------------+-----------
 EMP1003       |       14.67 |         1
 EMP1001       |       13.58 |         2
 EMP1005       |       13.42 |         3
 EMP1002       |       11.08 |         4
 EMP1007       |       10.00 |         5
 EMP1006       |        6.17 |         6
 EMP1004       |        5.08 |         7
(7 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

M4 - LAG - the rest period before each duty of every crew member

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- M4  LAG - the rest period before each duty of every crew member
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT cm.employee_code,
crewguard_submission-#        da.duty_start_utc,
crewguard_submission-#        da.duty_end_utc,
crewguard_submission-#        LAG(da.duty_end_utc) OVER (PARTITION BY da.crew_id ORDER BY da.duty_start_utc) AS previous_duty_end,
crewguard_submission-#        ROUND((EXTRACT(EPOCH FROM (da.duty_start_utc -
crewguard_submission(#               LAG(da.duty_end_utc) OVER (PARTITION BY da.crew_id ORDER BY da.duty_start_utc))) / 3600)::numeric, 2) AS rest_hours
crewguard_submission-# FROM duty_assignment da
crewguard_submission-# JOIN crew_member cm ON da.crew_id = cm.crew_id
crewguard_submission-# ORDER BY cm.employee_code, da.duty_start_utc;
 employee_code |      duty_start_utc       |       duty_end_utc        |     previous_duty_end     | rest_hours
---------------+---------------------------+---------------------------+---------------------------+------------
 EMP1001       | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |                           |
 EMP1001       | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 | 2026-09-01 09:45:00+05:30 |       0.25
 EMP1001       | 2026-09-03 06:00:00+05:30 | 2026-09-03 08:40:00+05:30 | 2026-09-01 14:15:00+05:30 |      39.75
 EMP1001       | 2026-09-03 08:45:00+05:30 | 2026-09-03 11:10:00+05:30 | 2026-09-03 08:40:00+05:30 |       0.08
 EMP1002       | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30 |                           |
 EMP1002       | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30 | 2026-09-02 09:45:00+05:30 |       0.08
 EMP1002       | 2026-09-06 06:30:00+05:30 | 2026-09-06 09:00:00+05:30 | 2026-09-02 12:45:00+05:30 |      89.75
 EMP1002       | 2026-09-06 09:05:00+05:30 | 2026-09-06 11:30:00+05:30 | 2026-09-06 09:00:00+05:30 |       0.08
 EMP1003       | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |                           |
 EMP1003       | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 | 2026-09-01 09:45:00+05:30 |       0.25
 EMP1003       | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30 | 2026-09-01 14:15:00+05:30 |      16.25
 EMP1003       | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30 | 2026-09-02 09:45:00+05:30 |       0.08
 EMP1004       | 2026-09-03 06:00:00+05:30 | 2026-09-03 08:40:00+05:30 |                           |
 EMP1004       | 2026-09-03 08:45:00+05:30 | 2026-09-03 11:10:00+05:30 | 2026-09-03 08:40:00+05:30 |       0.08
 EMP1005       | 2026-09-01 05:30:00+05:30 | 2026-09-01 09:45:00+05:30 |                           |
 EMP1005       | 2026-09-01 10:00:00+05:30 | 2026-09-01 14:15:00+05:30 | 2026-09-01 09:45:00+05:30 |       0.25
 EMP1005       | 2026-09-06 06:30:00+05:30 | 2026-09-06 09:00:00+05:30 | 2026-09-01 14:15:00+05:30 |     112.25
 EMP1005       | 2026-09-06 09:05:00+05:30 | 2026-09-06 11:30:00+05:30 | 2026-09-06 09:00:00+05:30 |       0.08
 EMP1006       | 2026-09-02 06:30:00+05:30 | 2026-09-02 09:45:00+05:30 |                           |
 EMP1006       | 2026-09-02 09:50:00+05:30 | 2026-09-02 12:45:00+05:30 | 2026-09-02 09:45:00+05:30 |       0.08
 EMP1007       | 2026-09-04 22:30:00+05:30 | 2026-09-05 03:30:00+05:30 |                           |
 EMP1007       | 2026-09-05 05:30:00+05:30 | 2026-09-05 10:30:00+05:30 | 2026-09-05 03:30:00+05:30 |       2.00
(22 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

M5 - Recursive CTE - duty count for every calendar day of a period

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- M5  Recursive CTE - duty count for every calendar day of a period
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# WITH RECURSIVE period(day) AS (
crewguard_submission(#     SELECT DATE '2026-09-01'
crewguard_submission(#     UNION ALL
crewguard_submission(#     SELECT day + 1 FROM period WHERE day < DATE '2026-09-07'
crewguard_submission(# )
crewguard_submission-# SELECT p.day,
crewguard_submission-#        COUNT(da.assignment_id) AS duties_that_day
crewguard_submission-# FROM period p
crewguard_submission-# LEFT JOIN duty_assignment da ON da.duty_start_utc::date = p.day
crewguard_submission-# GROUP BY p.day
crewguard_submission-# ORDER BY p.day;
    day     | duties_that_day
------------+-----------------
 2026-09-01 |               6
 2026-09-02 |               6
 2026-09-03 |               4
 2026-09-04 |               1
 2026-09-05 |               1
 2026-09-06 |               4
 2026-09-07 |               0
(7 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART N : FULL-TEXT SEARCH -

N0 - Setup: create index

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE INDEX IF NOT EXISTS idx_rule_fts
crewguard_submission-#     ON rule_definition
crewguard_submission-#     USING GIN (to_tsvector('english', rule_name || ' ' || description));
CREATE INDEX
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

N1 - Search the regulation catalogue

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- N1  Search the regulation catalogue
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT rule_code, rule_name, severity
crewguard_submission-# FROM rule_definition
crewguard_submission-# WHERE to_tsvector('english', rule_name || ' ' || description)
crewguard_submission-#       @@ plainto_tsquery('english', 'rest period');
 rule_code |          rule_name          | severity
-----------+-----------------------------+----------
 RST       | Minimum Rest Between Duties | HIGH
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT rule_code, rule_name
crewguard_submission-# FROM rule_definition
crewguard_submission-# WHERE to_tsvector('english', rule_name || ' ' || description)
crewguard_submission-#       @@ plainto_tsquery('english', 'certification aircraft');
 rule_code |             rule_name
-----------+-----------------------------------
 CER       | Valid Type Certification Required
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

N2 - Boolean full-text query with ranking

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- N2  Boolean full-text query with ranking
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT rule_code,
crewguard_submission-#        rule_name,
crewguard_submission-#        ROUND(ts_rank(to_tsvector('english', rule_name || ' ' || description),
crewguard_submission(#                      to_tsquery('english', 'duty & hours'))::numeric, 4) AS rank
crewguard_submission-# FROM rule_definition
crewguard_submission-# WHERE to_tsvector('english', rule_name || ' ' || description)
crewguard_submission-#       @@ to_tsquery('english', 'duty & hours')
crewguard_submission-# ORDER BY rank DESC;
 rule_code |               rule_name                |  rank
-----------+----------------------------------------+--------
 WDH       | Maximum Duty Hours in Seven Days       | 0.3188
 MDH       | Maximum Duty Hours in Twenty Eight Days | 0.3055
(2 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

N3 - Highlighted snippet of the matching regulation text

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- N3  Highlighted snippet of the matching regulation text
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# SELECT rule_code,
crewguard_submission-#        ts_headline('english', description, plainto_tsquery('english', 'rest')) AS matched_text
crewguard_submission-# FROM rule_definition
crewguard_submission-# WHERE to_tsvector('english', description) @@ plainto_tsquery('english', 'rest');
 rule_code |                               matched_text
-----------+-------------------------------------------------------------------------
 RST       | <b>rest</b> period between the end of one duty and the start of the next duty
(1 row)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

PART O : INDEXES AND QUERY PLAN -

O1 - Create indexes

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE INDEX IF NOT EXISTS idx_duty_crew_start  ON duty_assignment (crew_id, duty_start_utc);
CREATE INDEX
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE INDEX IF NOT EXISTS idx_duty_flight      ON duty_assignment (flight_id);
CREATE INDEX
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE INDEX IF NOT EXISTS idx_violation_run    ON violation (run_id);
CREATE INDEX
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE INDEX IF NOT EXISTS idx_violation_crew   ON violation (crew_id);
CREATE INDEX
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# CREATE INDEX IF NOT EXISTS idx_flight_dep_time  ON flight (dep_airport, sched_dep_utc);
CREATE INDEX
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

O2 - List the indexes (\di, commented in source line 1205)

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- List the indexes of the schema
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# \di
                              List of relations
 Schema |             Name              | Type  |  Owner   |        Table
--------+-------------------------------+-------+----------+----------------------
 public | aircraft_type_pkey            | index | postgres | aircraft_type
 public | airport_pkey                  | index | postgres | airport
 public | app_user_pkey                 | index | postgres | app_user
 public | app_user_username_key         | index | postgres | app_user
 public | certification_pkey            | index | postgres | certification
 public | crew_member_employee_code_key | index | postgres | crew_member
 public | crew_member_pkey              | index | postgres | crew_member
 public | duty_assignment_pkey          | index | postgres | duty_assignment
 public | excl_duty_no_overlap          | index | postgres | duty_assignment
 public | flight_pkey                   | index | postgres | flight
 public | idx_duty_crew_start           | index | postgres | duty_assignment
 public | idx_duty_flight               | index | postgres | duty_assignment
 public | idx_flight_dep_time           | index | postgres | flight
 public | idx_rule_fts                  | index | postgres | rule_definition
 public | idx_violation_crew            | index | postgres | violation
 public | idx_violation_run             | index | postgres | violation
 public | pk_violation_assignment       | index | postgres | violation_assignment
 public | rule_definition_pkey          | index | postgres | rule_definition
 public | uk_cert_crew_type_issue       | index | postgres | certification
 public | uk_duty_crew_flight           | index | postgres | duty_assignment
 public | uk_flight_number_dep          | index | postgres | flight
 public | validation_run_pkey           | index | postgres | validation_run
 public | violation_pkey                | index | postgres | violation
(23 rows)

PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

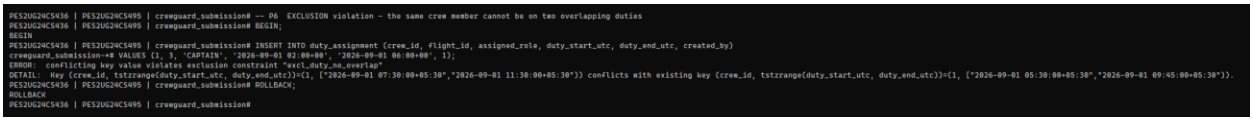
O3 - ANALYZE and EXPLAIN ANALYZE

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# ANALYZE;
ANALYZE
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# EXPLAIN ANALYZE
crewguard_submission-# SELECT da.assignment_id, da.duty_start_utc
crewguard_submission-# FROM duty_assignment da
crewguard_submission-# WHERE da.crew_id = 1
crewguard_submission-#   AND da.duty_start_utc >= '2026-09-01'
crewguard_submission-#   AND da.duty_start_utc <  '2026-09-08';
                                                
```

P5 - CHECK violation - a flight cannot depart from and arrive at the same airport

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- P5  CHECK violation - a flight cannot depart from and arrive at the same airport
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# BEGIN;
BEGIN
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO flight (flight_number, type_code, dep_airport, arr_airport, sched_dep_utc, sched_arr_utc)
crewguard_submission-*# VALUES ('CG998', 'A320', 'BLR', 'BLR', '2026-10-01 01:00+00', '2026-10-01 03:00+00');
ERROR:  new row for relation "flight" violates check constraint "chk_flight_distinct_endpoints"
DETAIL:  Failing row contains (12, CG998, A320, BLR, BLR, 2026-10-01 06:30:00+05:30, 2026-10-01 08:30:00+05:30).
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# ROLLBACK;
ROLLBACK
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

P6 - EXCLUSION violation - the same crew member cannot be on two overlapping duties



```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- P6  EXCLUSION violation - the same crew member cannot be on two overlapping duties
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# BEGIN;
BEGIN
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO duty_assignment (crew_id, flight_id, assigned_role, duty_start_utc, duty_end_utc, created_by)
crewguard_submission-*# VALUES (1, 3, 'CAPTAIN', '2026-09-01 02:00+00', '2026-09-01 06:00+00', 1);
ERROR:  conflicting key value violates exclusion constraint "excl_duty_no_overlap"
DETAIL:  Key (crew_id, tstzrange(duty_start_utc, duty_end_utc))=(1, ["2026-09-01 07:30:00+05:30","2026-09-01 11:30:00+05:30")) conflicts with existing key (crew_id, tstzrange(duty_start_utc, duty_end_utc))=(1, ["2026-09-01 05:30:00+05:30","2026-09-01 09:45:00+05:30")).
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# ROLLBACK;
ROLLBACK
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

P7 - CHECK violation - certificate expiry must be after the issue date

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- P7  CHECK violation - certificate expiry must be after the issue date
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# BEGIN;
BEGIN
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO certification (crew_id, type_code, issued_on, expires_on)
crewguard_submission-*# VALUES (1, 'B738', '2026-01-01', '2025-01-01');
ERROR:  new row for relation "certification" violates check constraint "chk_cert_validity_order"
DETAIL:  Failing row contains (9, 1, B738, 2026-01-01, 2025-01-01).
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# ROLLBACK;
ROLLBACK
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```

P8 - UNIQUE violation - a crew member cannot be assigned twice to one flight

```
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# -- P8  UNIQUE violation - a crew member cannot be assigned twice to one flight
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# BEGIN;
BEGIN
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# INSERT INTO duty_assignment (crew_id, flight_id, assigned_role, duty_start_utc, duty_end_utc, created_by)
crewguard_submission-*# VALUES (1, 1, 'CAPTAIN', '2026-09-20 00:00+00', '2026-09-20 04:00+00', 1);
ERROR:  duplicate key value violates unique constraint "uk_duty_crew_flight"
DETAIL:  Key (crew_id, flight_id)=(1, 1) already exists.
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission# ROLLBACK;
ROLLBACK
PES2UG24CS436 | PES2UG24CS495 | crewguard_submission#
```